A New Auxiliary Model Approach To Fermionic Criticality: Insights on Mottness in Strongly Correlated Systems

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by

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Declaration

Certificate by supervisor

From You, 5 Years Ago

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List Of Publications

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions. (Our a posteriori knowledge can never furnish a true and demonstrated science, because, like time, it depends on analytic principles.) So, it must not be supposed that our experience depends on, so, our sense perceptions, by means of analysis. Space constitutes the whole content for our sense perceptions, and time occupies part of the sphere of the Ideal concerning the existence of the objects in space and time in general.

Abstract

This thesis reports (i) the development of a new auxiliary model method for studying translation-invariant models of strongly-interacting electrons, and (ii) an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with emphasis on the signatures at a critical Fermi surface. Together, these results shed light on how a system of electrons behaves as it undergoes a correlation-driven metal-insulator transition on a lattice.

The auxiliary model method allows us to create a mapping between a lattice model and an auxiliary impurity model in terms of the Hamiltonian, eigenstates, Greens functions and entanglement measures, among other things. This mapping ensures that such lattice model quantities can be computed indirectly from similar computations within the auxiliary impurity model. In terms of phenomenology, this mapping connects various renormalisation group (RG) fixed points of the impurity model to those of the lattice model. As examples, local Fermi liquid and local moment phases of the impurity site get mapped to Fermi liquid and Mott insulating phases, respectively, on the lattice model. The mapping makes use of translation operators and a manybody version of Bloch's theorem to recreate the lattice model via repeated "tiling" operations on the impurity model. In contrast to other auxiliary model methods such as dynamical mean-field theory (DMFT) and its cluster variants (C-DMFT), this approach does not involve a self-consistency loop, making it more transparent. It also provides significantly more $k-$space resolution which is necessary for studying phenomena such as the pseudogap observed in the copper-oxide high-T_c superconductors.

A short description of the protocol for applying our method is as follows. We first identify the appropriate quantum impurity model corresponding to the lattice model at hand. For example, the impurity model must have a localisation-delocalisation transition on the impurity site in order to capture a metal-insulator transition on the lattice model. As another example, topological features of the band structure can be encoded by working with a topologically non-trivial conduction bath within the impurity model. Having decided upon a quantum impurity model, we solve the model using an impurity solver. In the works described in this thesis, we use the unitary renormalisation group because of its (i) analytical transparency, (ii) ability to deal with weak and strong correlations, (iii) ability to work at zero temperature, and (iv) reduced computational cost. Using the URG, we obtain a hierarchy of Hamiltonians that describe the high-energy and low-energy behaviour of the auxiliary model in various parameter regimes. This requires numerically solving the URG flow equations for various Hamiltonian couplings.

The phases of the impurity model are then characterised by (a) effective Hamiltonians, (b) static equal-time correlations, (c) dynamical correlations such as Greens functions and self-energy, and

(d) entanglement measures. Calculating some of these objects requires carrying out an iterative diagonalisation procedure on the hierarchy of Hamiltonians. Our tiling method then "periodises" these objects into the lattice model, generating required $k-$dependence in the process. Since the conduction bath of the impurity model already inherits the lattice geometry of the parent model, the appropriate symmetries are baked into the quantities. As an example, the lattice $k-$space Greens function G_k obtains contributions from the impurity site Greens function as well as impurity-bath off-diagonal Greens functions. While the former captures the local component of G_k, the latter captures the non-local components. With similar mappings existing for the dynamical correlations and entanglement measures, our method allows us to characterise various phases of the lattice model purely through impurity model computations.

Within our work on the holographic principle, we demonstrate the emergence of a holographic dimension in a system of 2D non-interacting Dirac fermions placed on a torus, by studying the scaling of multipartite entanglement measures under a sequence of renormalisation group (RG) transformations applied in momentum space. Geometric measures defined in this emergent space can be related to the RG beta function of the spectral gap, hence establishing a holographic connection between the spatial geometry of the emergent spatial dimension and the entanglement properties of the boundary quantum theory. We prove, analytically, that changing the boundedness of the holographic space involves a topological transition accompanied by a critical Fermi surface in the boundary theory. We go on to show that this results in the formation of a quantum wormhole geometry that connects the UV and the IR of the emergent dimension. The additional conformal symmetry at the transition also supports a relation between the emergent metric and the stress-energy tensor. In the presence of an Aharonov-Bohm flux, the entanglement gains a geometry-independent piece which is shown to be topological, sensitive to changes in boundary conditions, and related to the Luttinger volume of the system. Upon the insertion of a strong transverse magnetic field, we show that the Luttinger volume is linked to the Chern number of the occupied single-particle Landau levels.

Some Important Questions In this thesis, we have used the auxiliary model tiling method to study three fundamental problems in quantum condensed matter physics: (i) Mott metal-insulator transition on the Bethe lattice in infinite dimensions, (ii) Mott metal-insulator transition on the 2D square lattice, and (ii) quantum criticality in heavy fermion materials in 2D. Towards this, we have applied the method to two paradigmatic models: (a) an extended Hubbard model, and (ii) a bilayer extended Hubbard model. In preparation for applying the tiling approach, we have first analysed the corresponding quantum impurity models.

1. The Mott transition on the Bethe lattice is studied through an extended Anderson impurity model (eSIAM) that hosts local electronic correlation in the conduction bath proximate to the impurity site. Due to the lack of non-local correlations in this limit, it is sufficient to treat the conduction electron in the continuum limit.

2. The Mott transition in 2D is highly sensitive to details of the lattice. This requires us to study a lattice-embedded impurity model, where the impurity site is embedded in a 2D lattice of conduction electrons, and its hybridisation with the bath has the symmetries of the underlying lattice.

3. The heavy fermions problem necessitates a two layer impurity model, each layer acting as a lattice-embedded impurity model and coupled via inter-layer hybridisation.

The tiling approach then translates results from the auxiliary models into appropriate lattice models. In this thesis, we have attempted to answer a number of important open questions in the field:

- *Is there a minimal but effective quantum impurity model Hamiltonian that describes the Mott MIT of the* $1/2-$*filled Hubbard model on the Bethe lattice in infinite dimensions?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches. In the same context, is it possible to obtain a low-energy theory for the local gapless excitations precisely at the transition, where the metal is on the brink of destruction?

- For strongly-correlated electronic models, *is it possible to map the renormalisation group fixed points of such lattice models to those of quantum impurity models?* This would provide a firmer theoretical backing to various quantum impurity models, as well as reduce computational costs.

- The phenomenon of Mott insulation involves the localization of itinerant electrons due to strong local repulsion. Upon doping, a pseudogap (PG) phase emerges - marked by selective gapping of the Fermi surface without conventional symmetry breaking in spin or charge channels. *A key challenge is understanding how quasiparticle breakdown in the Fermi liquid gives rise to this enigmatic state, and how it connects to both the Mott insulating and superconducting phases.*

Questions on entanglement and holography become particularly interesting in systems of critical fermions due to the presence of (i) longer-ranged entanglement, (ii) topological features arising from the incompressibility of the Fermi volume and (iii) Fermi surface gapping phase transitions driven by changes in topological quantum numbers. Unlike topologically ordered phases, there is no geometry-independent topological term in the entanglement entropy of free fermionic systems. It is, however, possible to generate such a term even here by placing the system on a multiply connected manifold such as a cylinder, and inserting a magnetic flux through the cylinder. Several questions are therefore pertinent to this discussion:

- Not much is known about the holographic implications of a Fermi surface gapping transition. *What effect does a topological change in a Fermi surface have on the additional dimension generated for the system?* Can the transition be captured by topological invariants defined within the holographic space?

- *What happens to the topological signatures of entanglement entropy across a Fermi surface instability?* If the insulator on the other side has topological features, can we relate topological invariants on either side?

Summary Of The Chapters. For the readers' convenience, we now briefly describe the content of the various chapters.

The *Introduction* introduces the core themes of electronic correlation and Mott transition. After giving some preliminary insights from toy models, we introduce the other major theme – auxiliary models, and describe why quantum impurity models act as auxiliary models for models with strong local correlations. We go on to describe several classes of strongly correlated materials that are likely to be described by such methods, and discuss the phenomenology of some interacting models that will be investigated within the thesis. Near the end, we describe some of the features of the new auxiliary model developed within the thesis and list some of the questions addressed by it. We end by introducing the holographic principle and list some questions addressed by our work.

After the introduction, the *Methods and Preliminaries* chapter describes the various techniques and some preliminary ideas employed throughout the thesis. The chapter begins by deriving the unitary renormalisation group formalism and describes it's various properties and a practical protocol for implementing it. That is followed by demonstrations of the unitary RG on three simple models - a star graph model of spins in a magnetic field, the two-channel Kondo model, and the problem of a particle in a periodic potential. Through these problems, various aspects of the method are clarified. In order to provide more insights, the chapter also details how the unitary RG method is related to other RG-based methods that rely on canonical transformations, such as Poor Man's scaling, Schrieffer-Wolff transformation and CUT (continuous unitary transformation) RG. Apart from the URG, the chapter also describes an iterative diagonalisation method that has been used within this thesis to compute various correlation functions and entanglement measures from the hierarchy of Hamiltonians obtained from the URG. Since various entanglement measures have been used throughout the later chapters, we have also added a section on various entanglement measures here - what they mean, what are their inter-relations, why they are important and how one can compute some of them. The chapter concludes by describing some of the other strategies that have been employed to reduce computational costs in various projects.

In Chapter 3, *Kondo Frustration via Charge Fluctuations: A Route to Mott Localisation in Infinite Dimensions*, we study our first impurity model, the extended Anderson model with local correlation U_b in the conduction bath. We demonstrate that this model hosts a local version of the Mott metal-insulator transition on the Bethe lattice in infinite dimensions, as seen from DMFT. At a critical value of the ratio of U_b and the impurity-bath hybridisation, the effective impurity model shows a transition from a Kondo screened phase into an unscreened local moment phase. For the case of attractive local bath correlations, the model sheds new light on several aspects of the DMFT phase diagram. For example, the T = 0 metal-to-insulator quantum phase transition (QPT) is preceded by an excited state QPT (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The $T = 0$ gapless excitations at the quantum critical point display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening. A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

In Chapter 4, *A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models*, we lay out the details of the auxiliary model method developed to map impurity models to lattice models. We first motivate the general idea behind auxiliary model methods. This is followed by a construction of the manybody translation operators. We use this, along with a manybody Bloch's theorem, to construct lattice models from quantum impurity models and relate their eigenstates. This then allows us to map correlations functions, self-energies and entanglement measures across models. In particular, we demonstrate how the presence of the conduction bath introduces $k-$space resolution in various quantities.

In Chapter 5, *Mott Criticality as the Confinement Transition of of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions*, we apply the auxiliary model tiling method to a lattice-embedded extended SIAM. Applying a many-body tiling scheme to the fixed-point impurity model uncovers a lattice model with electron interactions and Kondo physics. At half-filling, the interplay between Kondo screening and bath charge fluctuations in the impurity model leads to Fermi liquid breakdown. This reveals a pseudogap phase characterized by a non-Fermi liquid (the Mott metal) residing on nodal arcs, gapped antinodal regions of the Fermi surface (Luttinger surfaces), and an anomalous scaling of the electronic scattering rate with frequency. The eventual confinement of holon-doublon excitations of this exotic metal obtains a continuous transition into the Mott insulator. Our results identify the pseudogap as a distinct long-range entangled quantum phase, and offer a new route to Mott criticality beyond the paradigm of local quantum criticality. The phase appears to encode multipartite entanglement, as seen from a large value of the quantum Fisher information. The critical point at the pseudogap-Mott insulator boundary is found to be described by the exactly-solvable Hatsugai-Kohmoto model. The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [44–46]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green's function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations.

In Chapter 8, *Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation*, we ...

In Chapter 7, *Holography of Entanglement in 2D Free Fermions: Emergent Geometry and Fermionic Criticality*, we provide an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

We conclude in the final chapter by summarising the main results obtained within the thesis, in terms of the methods developed as well as insights obtained into the behaviour of various models. We describe the impact of these results on the field, the broad conclusions emerging from them and some open questions resulting from the work.

Contents

Chapter 1

A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models

1.1 The Philosophy of Auxiliary Models

In the previous chapter, we extended the Anderson impurity model to host a correlation-driven delocalisation-localisation transition on the impurity site, from a local Fermi liquid state to a local moment state. We demonstrated that this transition captures very well the phenomenology of the Hubbard dimension in infinite dimensions (as seen from, say, dynamical mean-field theory (DMFT)). In this chapter, we extend this approach by developing a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [1]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix} . \tag{1.1}$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow} , \tag{1.2}$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum_{\langle i,j \rangle,\sigma}' \left(c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}\right) + U \sum_i' n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} \left(c^\dagger_{l,\sigma} c_{i,\sigma} + \text{h.c.}\right) , \tag{1.3}$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l.

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.1.

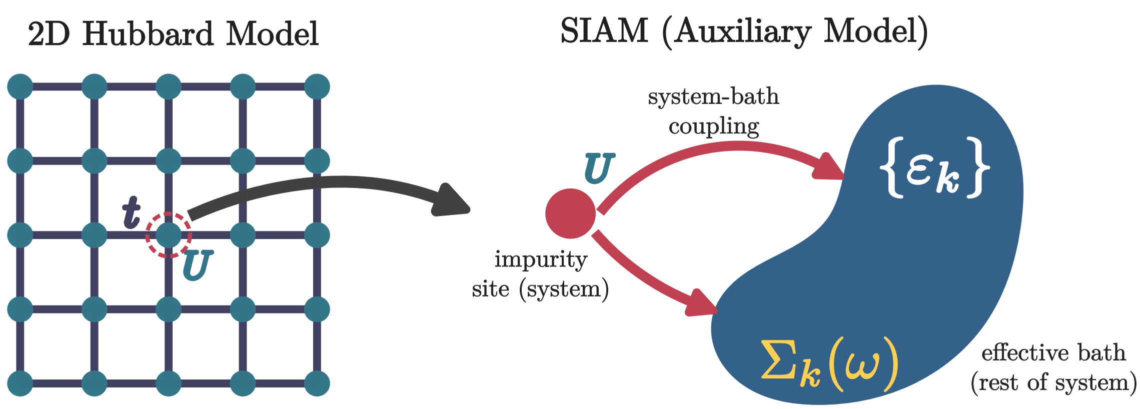

Figure 1.1: *Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H. Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N-1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V.*

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as $k-$space objects) will require a periodisation operation of an auxiliary model quantity.

The present work is aimed towards developing a new auxiliary model method to study correlated electronic systems. As described above, the first step is to identify the correct impurity model which can faithfully capture the physics of the lattice model we are trying to solve. The local behaviour of this impurity model should reflect the essential local physics of the lattice model. Typically, we will consider impurity model geometries where the real-space bath site connected directly to the impurity hosts some local interaction. We will henceforth refer to this site as the *zeroth site*. This model must be solved using an impurity solver. In the present work, we use the recently-developed unitary renormalisation group method [2–7]. The leap to the bulk model is then made by applying **manybody lattice translation operators** on the auxiliary model. This process, referred to as *tiling* here, allows us to reconstruct the lattice model Hamiltonian from that of the auxiliary model and restores the translation invariance of the full lattice model. Once the Hamiltonians can be mapped onto one another, it is possible to relate the eigenstates and correlations functions between the auxiliary model and lattice model, using a manybody version of **Bloch's theorem**.

1.2 What Is The Correct Auxiliary Model For Correlated Electrons on a 2D Square Lattice?

As discussed in more detail in the previous chapter, it was necessary to extend the single-impurity Anderson model (SIAM) in order to obtain a phase transition on the impurity site. Without any modification, the standard Anderson model (at particle-hole symmetry) leads to the emergence of an $\omega = 0$ Kondo resonance in the impurity ($d\sigma$) spectral function at any value of the impurity correlation U_d, at sufficiently low temperatures:

$$H_{\text{SIAM}} = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} - V \sum_{\sigma} (c^\dagger_{0,\sigma} c_{d,\sigma} + \text{h.c.}) + \varepsilon_d \sum_{\sigma} n_{d,\sigma} + U_d n_{d,\uparrow} n_{d,\downarrow} \, . \tag{2.1}$$

The operator $c_{0,\sigma}$ acts on the local conduction bath density that couples with the impurity site. The low-energy is one of local Fermi liquid for all parameter regimes. Increasing U_d only serves to reduce the width of the central peak at the cost of the appearance of side bands at energy scales of the order of U_d, but the resonance never disappears. The extended Anderson model that we developed augments the SIAM by allowing for local correlation in the local conduction bath charge density that couples with the impurity spin:

$$H_{\text{ESIAM}} = H_{\text{SIAM}} + J_K \boldsymbol{S}_d \cdot \boldsymbol{S}_0 - \frac{U_b}{2} \left(n_{0,\uparrow}^2 - n_{0,\downarrow}^2 \right) ; \tag{2.2}$$

tuning this bath correlation to large attractive values (see [8] for some recent findings of non-local effective attractive interactions within the Hubbard model) *does* lead to a localisation transition on the impurity where the local Fermi liquid excitations are replaced by the physics of a gapped local moment on the impurity. We note that the generation of such correlations in the bath is something that occurs in DMFT as well – the conduction bath acquires a *non-trivial self-energy* during the convergence to self-consistency.

While this modified impurity model captures well the Mott metal-insulator transition of the Hubbard model in infinite dimensions, it *lacks any knowledge of the underlying lattice geometry*, and is therefore ill-suited to address the phenomenology of more realistic low-dimensional models. Examples of such phenomenology include the pseudogap phase of the cuprates (with nodal-antinodal dichotomy), strange metal (with $T-$linear resistivity) and high-T_c superconducting phases observed in several families of quantum materials as well as graphene. This leads us to propose the framework of *lattice-embedded quantum impurity models*. Upon embedding the impurity site within the underlying lattice, the coupling between it and its environment must be consistent with the lowered symmetry of the lattice (see Fig. 1.2). Impurity-bath correlations calculated within such an impurity model are sensitive to the lattice geometry, and are much better suited to address the phenomenology mentioned above.

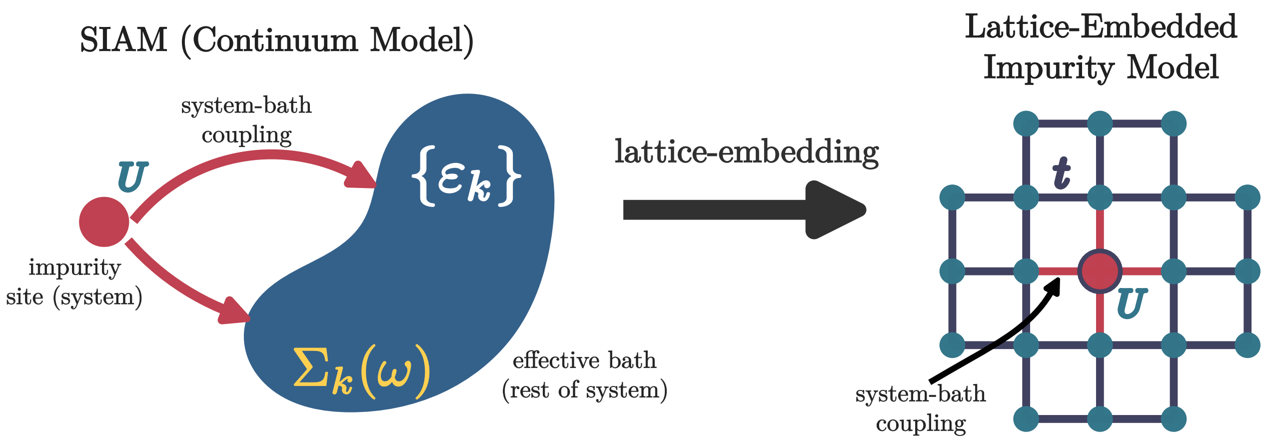

Figure 1.2

As a concrete example, we take the extended SIAM defined in eq. 2.2, and write down its lattice-embedded form on a 2D square lattice, with the impurity placed at the site r_d. This is done by allowing the impurity to hybridise with the four nearest-neighbours, through V and J_K. The same four sites also carry the local bath interaction. The complete auxiliary model is

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + \sum_{\langle i, r_d\rangle} \left[-\frac{V}{Z} \sum_\sigma (c^\dagger_{r_d,\sigma} c_{i,\sigma} + \text{h.c.}) + \frac{J_K}{Z} \boldsymbol{S}_{r_d} \cdot \boldsymbol{S}_i - \frac{W}{2Z} \left(n^2_{i,\uparrow} - n^2_{i,\downarrow} \right) \right] + \varepsilon_d \sum_\sigma n_{\boldsymbol{r}_d,\sigma} + U_d n_{\boldsymbol{r}_d,\uparrow} n_{\boldsymbol{r}_d,\downarrow}, \tag{2.3}$$

where Z is the coordination number of the lattice, and the second summation $\sum_{\langle i, r_d\rangle}$ is carried out over all sites i that are nearest-neighbour to the site r_d. Note that the one-particle hybridisation V and the spin-exchange coupling J_K have their *symmetry lowered* from the $s-$wave spherical symmetry of the continuum impurity models to a C_4-symmetry consistent with the underlying lattice. This feature is crucial in explaining the phenomenology of the various low-dimensional materials.

Various enhancements can be applied to such a model. The lattice can be modified to study different materials. More complicated geometries such as a two-layer model can be studied using a two-impurity auxiliary model, where each impurity couples with it's own conduction bath as well as to each other.

1.3 Mapping Auxiliary Model To Lattice Model

In this subsection, we provide an explicit example of constructing a lattice model from a lattice-embedded impurity model. This will involve the use of manybody lattice translation operators, so we begin with a brief discussion of them.

1.3.1 Manybody Translation Operators

In order to create the bulk model, we need to translate the local auxiliary model over the entire lattice. For this, we define *many-particle global* translation operators $T(\boldsymbol{a})$ that translate all positions by a vector $\boldsymbol{a}$. In terms of manybody states and operators, their action is defined as

$$\begin{aligned} T^\dagger(\boldsymbol{a}) \left|\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right\rangle &= \left|\boldsymbol{r}_1 + \boldsymbol{a}\right\rangle \otimes \left|\boldsymbol{r}_2 + \boldsymbol{a}\right\rangle \ldots \otimes \left|\boldsymbol{r}_n + \boldsymbol{a}\right\rangle \\ T^\dagger(\boldsymbol{a}) O\left(\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right) T(\boldsymbol{a}) &= O\left(\boldsymbol{r}_1 + \boldsymbol{a}, \boldsymbol{r}_2 + \boldsymbol{a}, \ldots\right) \ , \end{aligned} \tag{3.1}$$

where $\left|\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right\rangle$ is a state in the manyparticle Fock-space basis with the particles localised at the specified positions. For example, for a local fermionic creation operator $c^\dagger(\boldsymbol{r})$, we have

$$T^\dagger(\boldsymbol{a}) c^\dagger(\boldsymbol{r}) T(\boldsymbol{a}) = c^\dagger(\boldsymbol{r} + \boldsymbol{a}) \ . \tag{3.2}$$

It acts similarly on the auxiliary model Hamiltonian:

$$T^\dagger(\boldsymbol{a}) H_{\text{aux}}(\boldsymbol{r}_d) T(\boldsymbol{a}) = H_{\text{aux}}(\boldsymbol{r}_d + \boldsymbol{a}) \ , \tag{3.3}$$

translating all sites by the vector $\boldsymbol{a}$. By introducing the Fourier transform to momentum space,

$$\left|\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots\right\rangle = \otimes_{j=1}^N \int d\boldsymbol{k}_j e^{i\boldsymbol{r}_j \cdot \boldsymbol{k}_j} \left|\boldsymbol{k}_j\right\rangle \ , \tag{3.4}$$

it is easy to see that the total momentum states are eigenstates of the global translation operators:

$$\begin{aligned} T^\dagger(\boldsymbol{a}) \left|\boldsymbol{k}_1, \boldsymbol{k}_2, \ldots\right\rangle &= \otimes_{j=1}^N \int d\boldsymbol{r}_j e^{-i\boldsymbol{r}_j \cdot \boldsymbol{k}_j} T^\dagger(\boldsymbol{a}) \left|\boldsymbol{r}_j\right\rangle \ , \\ &= \otimes_{j=1}^N \int d\boldsymbol{r}_j e^{-i\boldsymbol{r}_j \cdot \boldsymbol{k}_j} \left|\boldsymbol{r}_j + \boldsymbol{a}\right\rangle \\ &= \otimes_{j=1}^N e^{i\boldsymbol{a} \cdot \boldsymbol{k}_j} \int d\boldsymbol{r}_j e^{-i\boldsymbol{r}_j \cdot \boldsymbol{k}_j} \left|\boldsymbol{r}_j\right\rangle \\ &= e^{i\boldsymbol{a} \cdot \boldsymbol{k}_{\text{tot}}} \left|\boldsymbol{k}_1, \boldsymbol{k}_2, \ldots\right\rangle , \end{aligned} \tag{3.5}$$

where $\boldsymbol{k}_{\text{tot}} = \sum_j \boldsymbol{k}_j$ is the total momentum. This gives a compact form for the translation operator:

$$T^\dagger(\boldsymbol{a}) = e^{i\boldsymbol{a} \cdot \hat{\boldsymbol{k}}_{\text{tot}}} \ , \tag{3.6}$$

where $\hat{\boldsymbol{k}}_{\text{tot}}$ is the total momentum operator.

We end this section by calculating the effect of the translation operators on the $k-$space annihilation operators. We have

$$c(\boldsymbol{k}) = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} c(\boldsymbol{r}) \ . \tag{3.7}$$

Applying the translation operator to this decomposition, we get

$$\begin{aligned} T\left(\vec{a}\right) c(\boldsymbol{k}) T^\dagger\left(\vec{a}\right) &= \frac{1}{\sqrt{N}}\sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} T\left(\vec{a}\right) c(\boldsymbol{r}) T^\dagger\left(\vec{a}\right) \\ &= \frac{1}{\sqrt{N}}\sum_{\boldsymbol{r}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}} c(\boldsymbol{r}-\vec{a}) \\ &= e^{-i\boldsymbol{k}\cdot\vec{a}} c(\boldsymbol{k}) \ . \end{aligned} \tag{3.8}$$

1.3.2 Periodisation of Auxiliary Model Into Extended Hubbard Model

We recall the lattice-embedded impurity model defined in eq. 2.3. As discussed above, the route to restoring translation symmetry and reconstructing the lattice model is to translate the auxiliary model using manybody translation operators. Accordingly, we identify the "unit cell" of translation. Since we expect the impurity site to capture most of the physics arising from the correlated electrons, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. In order to retain information of the lattice geometry, we convert the hybridisation and bath interaction terms from real space to $k-$space. The Hamiltonian for the unit cell is therefore

$$\begin{aligned} H_{\text{clust}}(\boldsymbol{r}_d) = &-\sum_{\boldsymbol{k},\sigma} e^{i\boldsymbol{k}\cdot\boldsymbol{r}_d} V(\boldsymbol{k}) (c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{k},\sigma} + \text{h.c.}) + \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}_d} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2) \boldsymbol{S}_{\boldsymbol{r}_d}\cdot\boldsymbol{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} + \varepsilon_d n_{\boldsymbol{r}_d} \\ &+ U_d n_{\boldsymbol{r}_d,\uparrow} n_{\boldsymbol{r}_d,\downarrow} - \frac{1}{2}\sum_{\{\boldsymbol{k}_i\},\sigma,\sigma'} \sigma\ \sigma'\ e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}_d} W(\{\boldsymbol{k}_i\}) c^\dagger_{\boldsymbol{k}_1,\sigma} c_{\boldsymbol{k}_2,\sigma} c^\dagger_{\boldsymbol{k}_3,\sigma'} c_{\boldsymbol{k}_4,\sigma'} \ , \end{aligned} \tag{3.9}$$

where $\boldsymbol{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} = \sum_{\alpha,\beta} \boldsymbol{\sigma}_{\alpha,\beta} c^\dagger_{\boldsymbol{k}_1,\alpha} c_{\boldsymbol{k}_2,\beta}$, and the Fourier transforms are defined by the following relations:

$$\begin{aligned} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}_d} V(\boldsymbol{k}) c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{k},\sigma} &= \frac{V}{Z}\sum_{r\to\text{NN of }\boldsymbol{r}_d} c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{r},\sigma} \ , \\ \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}_d} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2) \vec{S}_{\boldsymbol{r}_d}\cdot\vec{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} &= \frac{J_K}{Z}\sum_{r\to\text{NN of }\boldsymbol{r}_d} \boldsymbol{S}_{\boldsymbol{r}_d}\cdot\boldsymbol{S}_{\boldsymbol{r}} \ , \\ \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{q}_1,\boldsymbol{q}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}_d} W(\{\boldsymbol{k}_i\}) c^\dagger_{\boldsymbol{k}_1,\sigma} c_{\boldsymbol{k}_2,\sigma} c^\dagger_{\boldsymbol{k}_3,\sigma'} c_{\boldsymbol{k}_4,\sigma'} &= \frac{W}{2Z}\sum_{r\to\text{NN of }\boldsymbol{r}_d} n_{\boldsymbol{r},\sigma} n_{\boldsymbol{r},\sigma'} \ , \end{aligned} \tag{3.10}$$

We now translate this unit cell to all sites of the lattice, and reconstruct a translationally-invariant correlated lattice model:

$$H_{\text{latt}} = \sum_{\boldsymbol{r}} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d) H_{\text{clust}}(\boldsymbol{r}_d) T(\boldsymbol{r}-\boldsymbol{r}_d) \ . \tag{3.11}$$

We consider the effect of the translation operations on each part of the Hamiltonian. The purely local terms translate directly into a chemical potential term and a Hubbard interaction term:

$$\sum_{\boldsymbol{r}} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d)(\varepsilon_d \sum_\sigma n_{\boldsymbol{r}_d,\sigma} + U_d n_{\boldsymbol{r}_d,\uparrow} n_{\boldsymbol{r}_d,\downarrow}) T(\boldsymbol{r}-\boldsymbol{r}_d) = \varepsilon_d \sum_{\boldsymbol{r},\sigma} n_{\boldsymbol{r},\sigma} + U_d \sum_{\boldsymbol{r}} n_{\boldsymbol{r},\uparrow} n_{\boldsymbol{r},\downarrow} \ . \tag{3.12}$$

The one-particle hybridisation term maps onto a tight-binding hopping term:

$$\begin{aligned}\sum_{\boldsymbol{r}} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d) \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}_d} V(\boldsymbol{k}) c^\dagger_{\boldsymbol{r}_d,\sigma} c_{\boldsymbol{k},\sigma} T(\boldsymbol{r}-\boldsymbol{r}_d) &= \sum_{\boldsymbol{r}} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} V(\boldsymbol{k}) c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{k},\sigma} \\ &= \frac{V}{Z} \sum_{\boldsymbol{r}} \sum_{\boldsymbol{r}'\to \text{NN of } \boldsymbol{r}} c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{r}',\sigma} \\ &= \frac{2V}{Z} \sum_{\langle \boldsymbol{r},\boldsymbol{r}'\rangle} c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{r}',\sigma} \ ,\end{aligned} \tag{3.13}$$

where we used eq. 3.8 and the first equation from 3.10, and the fact that each nearest-neighbour pair $\boldsymbol{r}, \boldsymbol{r}'$ appears twice in the summation.

The Kondo terms map onto a nearest-neighbour spin-exchange term:

$$\begin{aligned}\sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d) e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}_d} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2) \vec{S}_{\boldsymbol{r}_d}\cdot\vec{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} T(\boldsymbol{r}-\boldsymbol{r}_d) &= \sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2)\cdot\boldsymbol{r}} J_K(\boldsymbol{k}_1,\boldsymbol{k}_2) \vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{k}_1,\boldsymbol{k}_2} \\ &= \frac{J_K}{Z} \sum_{\boldsymbol{r}} \sum_{\boldsymbol{r}'\to \text{NN of } \boldsymbol{r}} \vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{r}'} \ , \\ &= \frac{2J_K}{Z} \sum_{\langle \boldsymbol{r},\boldsymbol{r}'\rangle} \vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{r}'} \ ,\end{aligned} \tag{3.14}$$

where we used the second equation from 3.10. Finally, the bath interaction term W maps onto another Hubbard term

$$\begin{aligned}&\sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4} T^\dagger(\boldsymbol{r}-\boldsymbol{r}_d) e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}_d} W(\{\boldsymbol{k}_i\}) c^\dagger_{\boldsymbol{k}_1,\sigma} c_{\boldsymbol{k}_2,\sigma} c^\dagger_{\boldsymbol{k}_3,\sigma'} c_{\boldsymbol{k}_4,\sigma'} T(\boldsymbol{r}-\boldsymbol{r}_d) \\ &= \sum_{\boldsymbol{r},\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4} e^{i(\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3-\boldsymbol{k}_4)\cdot\boldsymbol{r}} W(\{\boldsymbol{k}_i\}) c^\dagger_{\boldsymbol{k}_1,\sigma} c_{\boldsymbol{k}_2,\sigma} c^\dagger_{\boldsymbol{k}_3,\sigma'} c_{\boldsymbol{k}_4,\sigma'} \\ &= \frac{W}{2Z} \sum_{\boldsymbol{r}} \sum_{\boldsymbol{r}'\to \text{NN of } \boldsymbol{r}} n_{\boldsymbol{r}',\sigma} n_{\boldsymbol{r}',\sigma'} \\ &= \frac{W}{2} \sum_{\boldsymbol{r}} n_{\boldsymbol{r},\sigma} n_{\boldsymbol{r},\sigma'} \ ,\end{aligned} \tag{3.15}$$

where we used the fact that each point $\boldsymbol{r}'$ (which we relabel as r) appears Z times in the summation.

The complete lattice Hamiltonian, after applying the translation operators, is

$$H_{\text{latt}} = \varepsilon_d \sum_{\boldsymbol{r},\sigma} n_{\boldsymbol{r},\sigma} + U_d \sum_{\boldsymbol{r}} n_{\boldsymbol{r},\uparrow} n_{\boldsymbol{r},\downarrow} - \frac{2V}{Z} \sum_{\langle \boldsymbol{r},\boldsymbol{r}'\rangle} c^\dagger_{\boldsymbol{r},\sigma} c_{\boldsymbol{r}',\sigma} + \frac{2J_K}{Z} \sum_{\langle \boldsymbol{r},\boldsymbol{r}'\rangle} \vec{S}_{\boldsymbol{r}}\cdot\vec{S}_{\boldsymbol{r}'} - \frac{W}{2} \sum_{\boldsymbol{r}} \left(n_{\boldsymbol{r},\uparrow} - n_{\boldsymbol{r},\downarrow}\right)^2 \ . \tag{3.16}$$

This is clearly an extended Hubbard model, with both 1-particle and spin-exchange nearest-neighbour hybridisation. The general form of the model is

$$H_{\text{ex-Hub}} = -t \sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + J \sum_{\langle i,j\rangle} \boldsymbol{S}_i \cdot \boldsymbol{S}_j + U \sum_i n_{i,\uparrow} n_{i,\downarrow} - \mu \sum_{i,\sigma} n_{i,\sigma} \ , \tag{3.17}$$

such that the effective couplings resulting from the tiling operation is

$$t = \frac{2V}{Z},\ J = \frac{2J_K}{Z},\ U = U_d - W,\ \mu = -\varepsilon_d + \frac{W}{2} \ . \tag{3.18}$$

1.3.3 Form Of The Eigenstates: Bloch's Theorem

The tiled Hamiltonian defined in eq. 3.11 is manifestly symmetric under discrete translation, given that it's expressed in terms of the translation operators. The fact that the Hamiltonian H_{latt} commutes with the manybody translation operator implies that the total crystal momentum is a conserved quantity. We will now use this to write down a "tiling" relation between the eigenstates of the auxiliary model and those of the lattice model.

In the tight-binding approach to lattice problems, the full Hamiltonian is obtained by translating the dynamics of the localised orbitals at each site over the entire lattice. The full eigenstate $|\Psi\rangle$ is obtained by constructing liner combinations of the eigenstates $|\psi(i)\rangle$ of the local Hamiltonians. According to Bloch's theorem, $|\Psi\rangle$ is of the form $|\Psi_{n,\boldsymbol{K}}\rangle = \sum_i e^{i\boldsymbol{K}\cdot\boldsymbol{r}_i} |\psi_n(i)\rangle$, where $\boldsymbol{K}$ is the single-particle crystal momentum, n is the band-index that labels various states within the local orbital, and $\boldsymbol{r}_i$ sums over the positions of the local Hamiltonians. Bloch's theorem ensures that eigenstates satisfy the following relation under a translation operation by an arbitrary number of lattice spacings $n\boldsymbol{a}$:

$$T^\dagger(l\boldsymbol{a}) |\Psi_{n,\boldsymbol{K}}\rangle = \sum_i e^{i\boldsymbol{K}\cdot\boldsymbol{r}_i} |\psi_n(i+l)\rangle = e^{-il\boldsymbol{K}\cdot\boldsymbol{a}} |\Psi_{n,\boldsymbol{K}}\rangle \ . \tag{3.19}$$

As discussed earlier, the eigenstates $|\Psi_{n,\boldsymbol{K}}\rangle$ of the lattice Hamiltonian obtained using eq. 3.11 enjoy the discrete translation symmetry of the lattice, and therefore enjoy a *many-body* Bloch's theorem [9]. This means that the *local* eigenstates $|\psi_n(\boldsymbol{r}_d)\rangle$ (with the impurity located at an arbitrary position $\boldsymbol{r}_d$) of the unit cell auxiliary model Hamiltonian $H_{\text{clust}}(\boldsymbol{r}_d)$ defined in eq. 3.9 can be used to construct eigenstates of the lattice Hamiltonian. The index $n(=0,1,\ldots)$ in the subscript indicates that it is the n^{th} eigenstate of the auxiliary model. The following combination of the auxiliary model eigenstates satisfies a many-particle equivalent of Bloch's theorem [9]:

$$|\Psi_s\rangle \equiv |\Psi_{\boldsymbol{K},n}\rangle = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{r}_d} e^{i\boldsymbol{K}\cdot\boldsymbol{r}_d} |\psi_n(\boldsymbol{r}_d)\rangle \ , \tag{3.20}$$

where N is the total number of lattice sites and $\boldsymbol{r}_d$ is summed over all lattice spacings. The set of $n=0$ states form the lowest band in the spectrum of the lattice, while higher values of n produce the more energetic bands.

1.4 Properties of the Bloch states

1.4.1 Translation invariance

It is easy to verify that $\Psi_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right)$ transforms like Bloch functions under translation by a displacement $\boldsymbol{r}$:

$$\begin{aligned}\Psi_{\vec{k}}\left(\{\boldsymbol{r}_k+\boldsymbol{r}\}\right) = \frac{1}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_i,\boldsymbol{a}} e^{i\vec{k}\cdot\vec{R}_i}\psi_{\text{aux}}\left(\boldsymbol{r}_i-\boldsymbol{r},\boldsymbol{a}\right) &= \frac{1}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_j,\boldsymbol{a}} e^{i\vec{k}\cdot\left(\vec{R}_j+\boldsymbol{r}\right)}\psi_{\text{aux}}\left(\boldsymbol{r}_j,\boldsymbol{a}\right)\\ &= e^{i\vec{k}\cdot\vec{R}}\Psi_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right)\end{aligned} \tag{4.1}$$

In the last equation, we transformed $\boldsymbol{r}_i \to \boldsymbol{r}_j = \boldsymbol{r}_i - \boldsymbol{r}$. Note that the argument $\boldsymbol{a}$ does not change under a translation of the system, because that vector always represents the difference between the impurity lattice position and its nearest neighbours, irrespective of the absolute position of the impurity.

The wavefunction can be even brought into the familiar Bloch function form:

$$\begin{aligned}\Psi^n_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right) &= \sum_{\boldsymbol{r}_i,\boldsymbol{a}}\frac{e^{i\vec{k}\cdot\vec{R}_i}}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\psi_{\text{aux}}\left(\{\boldsymbol{r}_d-\boldsymbol{r}_i,\boldsymbol{r}_0-\boldsymbol{r}_i-\boldsymbol{a}\}\right)\\ &= \frac{e^{i\vec{k}\cdot\frac{1}{N}\sum_k\vec{r}_k}}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_i,\boldsymbol{a}} e^{-i\vec{k}\cdot\left(\frac{1}{N}\sum_k\boldsymbol{r}_k-\vec{R}_i\right)}\psi_{\text{aux}}\left(\{\boldsymbol{r}_d-\boldsymbol{r}_i,\boldsymbol{r}_0-\boldsymbol{r}_i-\boldsymbol{a}\}\right)\\ &= e^{i\vec{k}\cdot\vec{r}_{\text{COM}}}\eta_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right)\end{aligned} \tag{4.2}$$

where $\boldsymbol{r}_{\text{COM}} = \frac{1}{N}\sum_k\boldsymbol{r}_k$ is the center-of-mass coordinate and

$$\eta_{\vec{k}}\left(\{\boldsymbol{r}_k\}\right) = \frac{1}{\lambda_{|\vec{k}\rangle}\mathcal{Z}N}\sum_{\boldsymbol{r}_i} e^{-i\vec{k}\cdot\left(\frac{1}{N}\sum_k\boldsymbol{r}_k-\vec{R}_i\right)}\psi^n_{\text{aux}}\left(\{\boldsymbol{r}_k-\boldsymbol{r}_i\}\right)\ , \tag{4.3}$$

is the translation symmetric function. This form of the eigenstate allows the interpretation that tuning the Bloch momentum $\vec{k}$ corresponds to a translation of the center of mass of the system (or in this case, of the auxiliary models that comprise the system).

1.4.2 Orthonormality

It is straightforward to show that these states form an orthonormal basis. We start by writing down the inner product of two distinct such states:

$$\begin{aligned}\langle\Psi_{\vec{k}'}|\Psi_{\vec{k}}\rangle &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\langle\psi_{\text{aux}}\left(\boldsymbol{r}_j,\boldsymbol{a}'\right)|\psi_{\text{aux}}\left(\boldsymbol{r}_i,\boldsymbol{a}\right)\rangle\\ &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)T^\dagger\left(\boldsymbol{r}_0-\boldsymbol{r}_j\right)T\left(\boldsymbol{r}_0-\boldsymbol{r}_i\right)|\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\rangle\end{aligned} \tag{4.4}$$

At this point, we insert a complete basis of momentum eigenkets $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle\vec{q}|$ to resolve the translation operators:

$$\begin{aligned}\langle\Psi_{\vec{k}'}|\Psi_{\vec{k}}\rangle &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}',\vec{q}} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)} \langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right) T\left(\boldsymbol{r}_j-\boldsymbol{r}_i\right)|\vec{q}\rangle\langle\vec{q}|\,\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\rangle\\
&= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{a},\boldsymbol{a}',\vec{q}} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)} \langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right) e^{i\vec{q}\cdot\left(\boldsymbol{r}_j-\boldsymbol{r}_i\right)}|\vec{q}\rangle\langle\vec{q}|\,\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\rangle\\
&= \frac{1}{|\lambda_{|\vec{k}\rangle}|^2\mathcal{Z}^2}\delta_{\vec{k},\vec{k}'}\sum_{\boldsymbol{a},\boldsymbol{a}'}\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)|\vec{k}\rangle\langle\vec{k}|\,\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\rangle\end{aligned} \tag{4.5}$$

At the last step, we used the summation form of the Kronecker delta function: $\sum_{\boldsymbol{r}_i} e^{i\boldsymbol{r}_i\left(\vec{k}-\vec{q}\right)}\sum_{\boldsymbol{r}_j} e^{i\boldsymbol{r}_j\left(\vec{q}-\vec{k}'\right)} = N\delta_{\vec{k},\vec{q}}N\delta_{\vec{k}',\vec{q}}$. The final remaining step is to identify that the inner products inside the summation are actually independent of the direction $\boldsymbol{a},\boldsymbol{a}'$ of the connecting vector, and both are equal to the normalisation factor $\lambda_{|\vec{k}\rangle}$:

$$\langle\Psi_{\vec{k}'}|\Psi_{\vec{k}}\rangle = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2\mathcal{Z}^2}\delta_{\vec{k},\vec{k}'}\sum_{\boldsymbol{a},\boldsymbol{a}'}|\lambda_{|\vec{k}\rangle}|^2 = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2\mathcal{Z}^2}\delta_{\vec{k},\vec{k}'}w^2|\lambda_{|\vec{k}\rangle}|^2 = \delta_{\vec{k},\vec{k}'} \tag{4.6}$$

This concludes the proof of orthonormality.

1.4.3 Eigenstates

To demonstrate that these are indeed eigenstates of the bulk Hamiltonian, we will calculate the matrix elements of the full Hamiltonian between these states:

$$\begin{aligned}\langle\Psi_{\vec{k}'}|\,H_{\text{H-H}}\,|\Psi_{\vec{k}}\rangle &= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\langle\psi_{\text{aux}}\left(\boldsymbol{r}_j,\boldsymbol{a}'\right)|\,H_{\text{aux}}\left(\boldsymbol{r}_k\right)|\psi_{\text{aux}}\left(\boldsymbol{r}_i,\boldsymbol{a}\right)\rangle\\
&= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)|\,T^\dagger_{\boldsymbol{r}_0-\boldsymbol{r}_j}T_{\boldsymbol{r}_0-\boldsymbol{r}_k}H_{\text{aux}}\left(\boldsymbol{r}_0\right)T^\dagger_{\boldsymbol{r}_0-\boldsymbol{r}_k}T_{\boldsymbol{r}_0-\boldsymbol{r}_i}|\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\rangle\\
&= \frac{1}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2}\sum_{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\boldsymbol{a},\boldsymbol{a}'} e^{i\left(\vec{k}\cdot\vec{R}_i-\vec{k}'\cdot\vec{R}_j\right)}\langle\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}'\right)|\,T^\dagger_{\boldsymbol{r}_k-\boldsymbol{r}_j}H_{\text{aux}}\left(\boldsymbol{r}_0\right)T_{\boldsymbol{r}_k-\boldsymbol{r}_i}|\psi_{\text{aux}}\left(\boldsymbol{r}_0,\boldsymbol{a}\right)\rangle\end{aligned} \tag{4.7}$$

To resolve the translation operators, we will now insert the momentum eigenstates $|\vec{k}\rangle$ of the full model: $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle\vec{q}|$ in between the operators.

$$
\begin{aligned}
\langle\Psi_{\vec{k}'}| H_{\text{H-H}} |\Psi_{\vec{k}}\rangle &= \sum_{\substack{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\\ \boldsymbol{a},\boldsymbol{a}',\vec{q},\vec{q}'}} \frac{e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)}}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2} \langle\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a}')|\vec{q}'\rangle \langle\vec{q}'| T^\dagger_{\boldsymbol{r}_k-\boldsymbol{r}_j} H_{\text{aux}}(\boldsymbol{r}_0) T_{\boldsymbol{r}_k-\boldsymbol{r}_i}|\vec{q}\rangle \langle\vec{q}|\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a})\rangle \\
&= \sum_{\substack{\boldsymbol{r}_i,\boldsymbol{r}_j,\boldsymbol{r}_k,\\ \boldsymbol{a},\boldsymbol{a}',\vec{q},\vec{q}'}} \frac{e^{i\left(\vec{k}\cdot\vec{R}_i - \vec{k}'\cdot\vec{R}_j\right)}}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2N^2} \langle\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a}')|\vec{q}'\rangle e^{-i\vec{q}'\cdot(\boldsymbol{r}_k-\boldsymbol{r}_j)} e^{i\vec{q}\cdot(\boldsymbol{r}_k-\boldsymbol{r}_i)} \langle\vec{q}'|H_{\text{aux}}(\boldsymbol{r}_0)|\vec{q}\rangle \langle\vec{q}|\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a}
\\
&= \frac{N}{\lambda^*_{|\vec{k}'\rangle}\lambda_{|\vec{k}\rangle}\mathcal{Z}^2} \sum_{\boldsymbol{a},\boldsymbol{a}',\vec{q},\vec{q}'} \delta_{\vec{k},\vec{q}}\delta_{\vec{q},\vec{q}'}\delta_{\vec{q}',\vec{k}} \langle\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a}')|\vec{q}'\rangle \langle\vec{q}'|H_{\text{aux}}(\boldsymbol{r}_0)|\vec{q}\rangle \langle\vec{q}|\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a})\rangle \\
&= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2\mathcal{Z}^2}\delta_{kk'} \sum_{\boldsymbol{a},\boldsymbol{a}'} \langle\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a}')|\vec{k}\rangle \langle\vec{k}|H_{\text{aux}}(\boldsymbol{r}_0)|\vec{k}\rangle \langle\vec{k}|\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a})\rangle \\
&= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2\mathcal{Z}^2}\delta_{kk'} \langle\vec{k}|H_{\text{aux}}(\boldsymbol{r}_0)|\vec{k}\rangle \underbrace{\sum_{\boldsymbol{a},\boldsymbol{a}'} \langle\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a}')|\vec{k}\rangle \langle\vec{k}|\psi_{\text{aux}}(\boldsymbol{r}_0,\boldsymbol{a})\rangle}_{w^2|\lambda_{|\vec{k}\rangle}|^2} \\
&= N\delta_{kk'} \langle\vec{k}|H_{\text{aux}}(\boldsymbol{r}_0)|\vec{k}\rangle
\end{aligned}
\tag{4.8}
$$

The momentum eigenket $|\vec{k}\rangle$ can be expressed as a sum over position eigenkets at varying distances from $\boldsymbol{r}_0$. This form allows a systematic method of improving the energy eigenvalue estimate, because the interacting in the impurity model is extremely localised, so the overlap between two auxiliary models decreases rapidly with distance. The majority of the contribution will come from the state at $\boldsymbol{r}_0$, and improvements are then made by considering auxiliary models at further distances.

1.5 Mapping Greens Functions from Auxiliary Model to Lattice Model

1.5.1 One-particle $k-$space Greens functions

In the previous part, we proposed a form for the eigenstates $|\Psi_s\rangle$ of the bulk Hamiltonian in terms of the ground-states $|\psi_n\rangle$ of the auxiliary models. In this section, we will relate one-particle Greens functions of the lattice model to those of the auxiliary model. We define the retarded time-domain lattice $k-$space Greens function at zero temperature as

$$
G(\boldsymbol{k}\sigma;t) = -i\theta(t)\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\rangle\ . \tag{5.1}
$$

where $|\Psi_{\text{gs}}\rangle$ is the lattice groundstate. Time evolution of the operators is governed by the lattice Hamiltonian H_{latt}:

$$
c_{\boldsymbol{k}\sigma}(t) = e^{itH_{\text{latt}}}c_{\boldsymbol{k}\sigma}e^{-itH_{\text{latt}}}\ . \tag{5.2}
$$

By using eq. 3.8 and the fact that the translation operators commute with H_{latt}, we have the relation

$$T\left(\vec{a}\right)c(\boldsymbol{k})(t)T^\dagger\left(\vec{a}\right) = e^{-i\boldsymbol{k}\cdot\vec{a}}c(\boldsymbol{k})(t)\ , \tag{5.3}$$

which combined with eq. 3.8 gives

$$T\left(\boldsymbol{a}\right)c(\boldsymbol{k})(t)c^\dagger(\boldsymbol{k})T^\dagger\left(\boldsymbol{b}\right) = e^{i\boldsymbol{k}\cdot(\boldsymbol{b}-\boldsymbol{a})}c(\boldsymbol{k})(t)T^\dagger(\boldsymbol{b}-\boldsymbol{a})c^\dagger(\boldsymbol{k})\ . \tag{5.4}$$

In order to make use of the mapping between auxiliary model and lattice model Hamiltonians, we would like to express this lattice Greens function in terms of auxiliary model Greens functions – objects that can be computed purely from within a single auxiliary model. This will be a major simplification in our computations, because it is easier to work with a quantum impurity model, as compared to an interacting lattice model.

To make progress towards this, we consider the equivalent object

$$G(\boldsymbol{k}\sigma;t) = -i\theta(t)\frac{1}{2}\left[\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\rangle + \langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}, c^\dagger_{\boldsymbol{k}\sigma}(-t)\}|\Psi_{\text{gs}}\rangle\right]\ . \tag{5.5}$$

We consider only the first term for now. We express the lattice groundstate in terms of the auxiliary model groundstates (using eq. 3.20):

$$\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\Psi_{\text{gs}}\rangle = \frac{1}{N}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}\langle\psi_{\text{gs}}(\boldsymbol{r}_2)|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle\ . \tag{5.6}$$

We take one of the terms in the anticommutator and proceed to simplify it:

$$\begin{aligned}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}\langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|T(\boldsymbol{r}_2-\boldsymbol{r}_d)c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\\ &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}e^{i\boldsymbol{k}(\boldsymbol{r}_1-\boldsymbol{r}_2)}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_2)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\\ &= N\sum_{\boldsymbol{r}}e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)T^\dagger(\boldsymbol{r})c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\ .\end{aligned} \tag{5.7}$$

where $\boldsymbol{r}_d$ is a reference lattice site, and we used eq. 5.4 in the second step. The other term in the anticommutator gives

$$\begin{aligned}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}\langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c^\dagger_{\boldsymbol{k}\sigma}c_{\boldsymbol{k}\sigma}(t)|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|T(\boldsymbol{r}_2-\boldsymbol{r}_d)c^\dagger_{\boldsymbol{k}\sigma}c_{\boldsymbol{k}\sigma}(t)T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_d)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\\ &= \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2}e^{-i\boldsymbol{k}(\boldsymbol{r}_1-\boldsymbol{r}_2)}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c^\dagger_{\boldsymbol{k}\sigma}T^\dagger(\boldsymbol{r}_1-\boldsymbol{r}_2)c_{\boldsymbol{k}\sigma}(t)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\\ &= N\sum_{\boldsymbol{r}}e^{-i\boldsymbol{k}\cdot\boldsymbol{r}}\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c^\dagger_{\boldsymbol{k}\sigma}T^\dagger(\boldsymbol{r})c_{\boldsymbol{k}\sigma}(t)|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\ .\end{aligned} \tag{5.8}$$

To resolve the time evolution of the operators, we insert $1 = \sum_m |\psi_m(\boldsymbol{r}_d)\rangle \langle\psi_m(\boldsymbol{r}_d)|$ into the equation. We again work with just the first term of the anticommutator for convenience:

$$\begin{aligned}
&\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle \\
&= N\sum_{\boldsymbol{r},n_1,n_2} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)\,|\psi_{n_1}(\boldsymbol{r}_d)\rangle \langle\psi_{n_1}(\boldsymbol{r}_d)|\,T^\dagger(\boldsymbol{r})\,|\psi_{n_2}(\boldsymbol{r}_d)\rangle \langle\psi_{n_2}(\boldsymbol{r}_d)|\,c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\
&= N\sum_{\boldsymbol{r},n_1,n_2} e^{i\boldsymbol{k}\cdot\boldsymbol{r}} \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)\,|\psi_{n_1}(\boldsymbol{r}_d)\rangle \langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_2}(\boldsymbol{r}_d+\boldsymbol{r})\rangle \langle\psi_{n_2}(\boldsymbol{r}_d)|\,c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\ .
\end{aligned} \tag{5.9}$$

The overlap $\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_2}(\boldsymbol{r}_d+\boldsymbol{r})\rangle$ between the eigenstates of the auxiliary models at $\boldsymbol{r}_d$ and $\boldsymbol{r}_d+\boldsymbol{r}$ are appreciable only for $n_1 = n_2$. This is because, for small $\boldsymbol{r}$, the auxiliary models are very similar and the states are therefore almost orthogonal. For large $\boldsymbol{r}$, the overlap is again suppressed because the auxiliary model eigenstates are localised around the impurity. We therefore make the simplifying assumption: $\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_2}(\boldsymbol{r}_d+\boldsymbol{r})\rangle = \delta_{n_1,n_2}\langle\psi_{n_1}(\boldsymbol{r}_d)|\psi_{n_1}(\boldsymbol{r}_d+\boldsymbol{r})\rangle = \delta_{n_1,n_2}F_{n_1}(\boldsymbol{r})$. Substituting this, we get

$$\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2} \langle\psi_{\text{gs}}(\boldsymbol{r}_2)|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_1)\rangle = N\sum_n F_n(\boldsymbol{k})\,\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{k}\sigma}(t)\,|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|\,c^\dagger_{\boldsymbol{k}\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle\ , \tag{5.10}$$

where we have defined the Fourier transform $F_n(\boldsymbol{k}) = \sum_{\boldsymbol{r}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}F_n(\boldsymbol{r})$ of the overlap function.

We now consider more carefully the transition operator $c_{\boldsymbol{k}\sigma}$ for the 1-particle excitation giving rise to the above Greens function. Within our auxiliary model approach, gapless excitations within the lattice model are represented by gapless excitations of the impurity site, specifically those that try to screen the impurity site. As a result, the uncoordinated scattering process $c_{\boldsymbol{k}\sigma}$ for the lattice model must be replaced by a coherent excitation $\mathcal{T}$ within the impurity model that captures those gapless excitations that occur in connection with the impurity, and projects out the uncorrelated excitations that take place even when the impurity site is decoupled from the bath.

In order to construct this auxiliary model $\mathcal{T}-$matrix, we allow for all excitations of the impurity site in conjunction with those of the $k-$state:

$$\mathcal{T}_{\boldsymbol{k},\sigma} = \left[c^\dagger_{\boldsymbol{r}_d\sigma} + S^\sigma_{\boldsymbol{r}_d} + C^+_{\boldsymbol{r}_d}\right] c_{\boldsymbol{k}\sigma}\ , \tag{5.11}$$

where $S^+_{\boldsymbol{r}_d}$ and $C^+_{\boldsymbol{r}_d}$ are spin and charge flip operators on the impurity site that cause transition from $|\downarrow\rangle \to |\uparrow\rangle$ and $|0\rangle \to |\uparrow\downarrow\rangle$ respectively. The first term encodes the local component of the $k-$space Greens function by creating an excitation on the impurity site (which is then translated across the lattice), and the rest of the terms encode the non-local components by tracking impurity-bath excitations.

Apart from modifying the excitation operator, we will now simplify the time-evolution as well. The first overlap on the right-hand side of eq. 5.10 expands to

$$\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|e^{i(H_{\text{aux}}(\boldsymbol{r}_d)+H_{\text{cav}}(\boldsymbol{r}_d))t}\mathcal{T}_{\boldsymbol{k},\sigma}e^{-i(H_{\text{aux}}(\boldsymbol{r}_d)+H_{\text{cav}}(\boldsymbol{r}_d))t}|\psi_n(\boldsymbol{r}_d)\rangle\ , \tag{5.12}$$

where $H_{\rm aux}(\boldsymbol{r}_d)$ (eq. 2.3) is the auxiliary model Hamiltonian consisting of the cluster Hamiltonian $H_{\rm clust}(\boldsymbol{r}_d)$ along with all one-particle terms $H_{\rm 1p}$ in the Hamiltonian (in a typical lattice model, this will be he tight-binding hopping terms), and $H_{\rm cav}(\boldsymbol{r}_d)$ is the "cavity" Hamiltonian consisting of all interaction (2-particle and beyond) terms in the Hamiltonian. Together, they create the full lattice Hamiltonian: $H_{\rm latt} = H_{\rm aux}(\boldsymbol{r}_d) + H_{\rm cav}(\boldsymbol{r}_d)$. Using the Zassenhaus form of the Baker-Campbell-Hausdorff formula, we have

$$e^{i(H_{\rm aux}(\boldsymbol{r}_d)+H_{\rm cav}(\boldsymbol{r}_d))t} \approx e^{iH_{\rm aux}(\boldsymbol{r}_d)t}e^{iH_{\rm cav}(\boldsymbol{r}_d)t}e^{[H_{\rm aux}(\boldsymbol{r}_d),\, H_{\rm cav}(\boldsymbol{r}_d)]t^2/2} \; , \tag{5.13}$$

where we have dropped the higher-order terms because the commutators involve only the boundary degrees of freedom and become subsequently smaller as the order increases. One can think of the operator $e^{iH_{\rm cav}(\boldsymbol{r}_d)t}e^{[H_{\rm aux}(\boldsymbol{r}_d),\, H_{\rm cav}(\boldsymbol{r}_d)]t^2/2}$ as renormalising the auxiliary model $\mathcal{T}-$matrix through hybridisation with the rest of the system and formally leading to a new $\mathcal{T}-$matrix:

$$\mathcal{T}^*_{\boldsymbol{k},\sigma} = e^{iH_{\rm cav}(\boldsymbol{r}_d)t}e^{[H_{\rm aux}(\boldsymbol{r}_d),\, H_{\rm cav}(\boldsymbol{r}_d)]t^2/2}\mathcal{T}_{\boldsymbol{k},\sigma}e^{-[H_{\rm aux}(\boldsymbol{r}_d),\, H_{\rm cav}(\boldsymbol{r}_d)]t^2/2}e^{-iH_{\rm cav}(\boldsymbol{r}_d)t} \; . \tag{5.14}$$

One can also absorb the modulating phase factor $F_n(\boldsymbol{k})$ (defined above and appearing in eq. 5.10) into this renormalisation:

$$\mathcal{T}^*_{\boldsymbol{k},\sigma} = e^{iH_{\rm cav}(\boldsymbol{r}_d)t}e^{[H_{\rm aux}(\boldsymbol{r}_d),\, H_{\rm cav}(\boldsymbol{r}_d)]t^2/2}\mathcal{T}_{\boldsymbol{k},\sigma}F_n(\boldsymbol{k})\mathcal{P}_n(\boldsymbol{r}_d)\; e^{-[H_{\rm aux}(\boldsymbol{r}_d),\, H_{\rm cav}(\boldsymbol{r}_d)]t^2/2}e^{-iH_{\rm cav}(\boldsymbol{r}_d)t} \; , \tag{5.15}$$

where $\mathcal{P}_n(\boldsymbol{r}_d)$ projects onto the $n^{\rm th}$ eigenstate $|\psi_n(\boldsymbol{r}_d)\rangle$.

Since the overlap is calculated in the eigenstates of the auxiliary model at $\boldsymbol{r}_d$, we adopt the zeroth order approximation of ignoring this renormalisation. Later, we will specify a prescription of systematically improving our computation. The overlap expression becomes

$$\langle\psi_{\rm gs}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \langle\psi_{\rm gs}(\boldsymbol{r}_d)|e^{i(H_{\rm clust}(\boldsymbol{r}_d)+H_{\rm 1p})t}\mathcal{T}_{\boldsymbol{k},\sigma}e^{-i(H_{\rm clust}(\boldsymbol{r}_d)+H_{\rm 1p})t}|\psi_n(\boldsymbol{r}_d)\rangle \; . \tag{5.16}$$

To make the problem tractable, we treat the effect of $H_{\rm 1p}$ at the level of an expectation value and note that the effect of the transition operator is to create a hole of momentum $\boldsymbol{k}$ (eq. 5.11). Let the Hamiltonian $H_{\rm 1p}$ be diagonal in certain modes $c_{\nu,\sigma}$ with energies ϵ_ν, and the operator $c_{\boldsymbol{k},\sigma}$ can be expressed in terms of these modes: $c_{\boldsymbol{k},\sigma} = \sum_\nu C_{\nu,\boldsymbol{k}}c_{\nu,\sigma}$. Expanding the $\mathcal{T}-$matrix in this fashion gives

$$\langle\psi_{\rm gs}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \sum_\nu C_{\nu,\boldsymbol{k}}\, \langle\psi_{\rm gs}(\boldsymbol{r}_d)|e^{i(H_{\rm clust}(\boldsymbol{r}_d)+H_{\rm 1p})t}\mathcal{T}_{\nu,\sigma}e^{-i(H_{\rm clust}(\boldsymbol{r}_d)+H_{\rm 1p})t}|\psi_n(\boldsymbol{r}_d)\rangle \; . \tag{5.17}$$

Each excitation $\mathcal{T}_{\nu,\sigma}$ therefore leads to an energy cost $-\epsilon_\nu$. Moreover, $|\psi_n(\boldsymbol{r}_d)\rangle$ and $|\psi_{\rm gs}(\boldsymbol{r}_d)\rangle$ are eigenstates of $H_{\rm clust}(\boldsymbol{r}_d)$ with energies E_n and E_0. This leads to the following simplified expression:

$$\langle\psi_{\rm gs}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{k},\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle = \sum_\nu C_{\nu,\boldsymbol{k}}e^{-i(E_n-E_0+\epsilon_\nu)t}\, \langle\psi_{\rm gs}(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle \; . \tag{5.18}$$

Substituting the expression back into 5.10 and then into 5.6, we get an expression for one half of the first part of the retarded Greens function (the full form is in eq. 5.5):

$$\begin{aligned} &-i\theta(t)\,\langle\Psi_{\rm gs}|c_{\boldsymbol{k}\sigma}(t)c^\dagger_{\boldsymbol{k}\sigma}|\Psi_{\rm gs}\rangle \\ &= -i\theta(t)\sum_{\nu,n} C_{\nu,\boldsymbol{k}}e^{-i(E_n-E_0+\epsilon_\nu)t}\,\langle\psi_{\rm gs}(\boldsymbol{r}_d)|\mathcal{T}_{\nu,\sigma}|\psi_n(\boldsymbol{r}_d)\rangle\,\langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}|\psi_{\rm gs}(\boldsymbol{r}_d)\rangle \; . \end{aligned} \tag{5.19}$$

Working through the other term of eq. 5.6 with the same strategy, we get the full expression:

$$\begin{aligned}
&-i\theta(t)\left\langle \Psi_{\text{gs}}\right|\{c_{\boldsymbol{k}\sigma}(t), c^\dagger_{\boldsymbol{k}\sigma}\}\left|\Psi_{\text{gs}}\right\rangle \\
&= -i\theta(t)\sum_{\nu,n} C_{\nu,\boldsymbol{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}_{\nu,\sigma}\left|\psi_n(\boldsymbol{r}_d)\right\rangle \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle \\
&\quad - i\theta(t)\sum_{\nu,n} C_{\nu,\boldsymbol{k}} e^{i(E_n - E_0 - \epsilon_\nu)t} \left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}\left|\psi_n(\boldsymbol{r}_d)\right\rangle \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}_{\nu,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle \ .
\end{aligned} \tag{5.20}$$

Finally, we consider the other term in eq. 5.5, $-i\theta(t)\left\langle \Psi_{\text{gs}}\right|\{c_{\boldsymbol{k}\sigma}, c^\dagger_{\boldsymbol{k}\sigma}(-t)\}\left|\Psi_{\text{gs}}\right\rangle$. Proceeding in the same way, eq. 5.18 is replaced by

$$\begin{aligned}
\left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}(-t)\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle &= \left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}_{\boldsymbol{k},\sigma}(-t)\left|\psi_n(\boldsymbol{r}_d)\right\rangle^* \\
&= \sum_\nu C^*_{\nu,\boldsymbol{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\nu,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle \ .
\end{aligned} \tag{5.21}$$

Evaluating this term exactly as before and changing from time to frequency domain using $g(\omega) = \int dt\ e^{i\omega T} f(t)$ gives

$$\begin{aligned}
G(\boldsymbol{k},\sigma;\omega) = \sum_{\nu,n} C_{\nu,\boldsymbol{k}} \Bigg[&\frac{\left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}_{\nu,\sigma}\left|\psi_n(\boldsymbol{r}_d)\right\rangle \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle}{\omega - (E_n - E_0 + \epsilon_\nu)} \\
&+ \frac{\left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}\left|\psi_n(\boldsymbol{r}_d)\right\rangle \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}_{\nu,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle}{\omega + (E_n - E_0 - \epsilon_\nu)} \Bigg] + \text{h.c.}
\end{aligned} \tag{5.22}$$

We recognise the two parts of the lattice Greens function as a retarded off-diagonal Greens functions calculated within the auxiliary model at $\boldsymbol{r}_d$, but at shifted frequencies $\omega - \epsilon_\nu$:

$$G(\boldsymbol{k},\sigma;\omega) = \sum_\nu \left[C_{\nu,\boldsymbol{k}} \mathcal{G}(\mathcal{T}_{\nu,\sigma}, \mathcal{T}^\dagger_{\boldsymbol{k},\sigma}; \omega - \epsilon_\nu) + C^*_{\nu,\boldsymbol{k}} \mathcal{G}(\mathcal{T}_{\boldsymbol{k},\sigma}, \mathcal{T}^\dagger_{\nu,\sigma}; \omega - \epsilon_\nu) \right] \tag{5.23}$$

We now discuss how to systematically improve this estimate. One way is to include the effect of other auxiliary models in the overlap function $F_n(\boldsymbol{r})$. This will of course require us to calculate an off-diagonal Greens function. As a concrete example, upon including the effects of the auxiliary models nearest to the reference point $\boldsymbol{r}_d$, the expression for the Greens function becomes

$$\begin{aligned}
G(\boldsymbol{k},\sigma;\omega) = \sum_{\nu,n} C_{\nu,\boldsymbol{k}} \Bigg[&\frac{\left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}^*_{\nu,\sigma}\left|\psi_n(\boldsymbol{r}_d)\right\rangle \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle}{\omega - \epsilon_\nu - (E_n - E_0)} \\
&+ \frac{\left\langle \psi_{\text{gs}}(\boldsymbol{r}_d)\right|\mathcal{T}^\dagger_{\boldsymbol{k},\sigma}\left|\psi_n(\boldsymbol{r}_d)\right\rangle \left\langle \psi_n(\boldsymbol{r}_d)\right|\mathcal{T}^*_{\nu,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle}{\omega - \epsilon_\nu + (E_n - E_0)} \Bigg] \ ,
\end{aligned} \tag{5.24}$$

with $\tau^*_{\nu,\sigma} = \mathcal{T}_{\nu,\sigma}(F_n(0) + \sum_{\boldsymbol{a}} F_n(\boldsymbol{a}) e^{i\boldsymbol{k}\cdot\boldsymbol{a}})$ where $\boldsymbol{a}$ sums over the primitive lattice vectors. Another way to include more non-local contributions is by considering the effects of the cavity Hamiltonian in eq. 5.15:

$$\mathcal{T}^*_{\boldsymbol{k},\sigma}(t) = e^{iH_{\text{cav}}(\boldsymbol{r}_d)t}(\mathcal{T}_{\boldsymbol{k},\sigma} + t^2/2\left[[H_{\text{aux}}(\boldsymbol{r}_d),\ H_{\text{cav}}(\boldsymbol{r}_d)], \mathcal{T}_{\boldsymbol{k},\sigma} \right]) e^{-iH_{\text{cav}}(\boldsymbol{r}_d)t} \ . \tag{5.25}$$

The time-domain Greens function associated with this operator will then need to be calculated, which can then be used to calculate frequency-domain functions.

1.5.2 One-particle real-space Greens functions

We now work with Greens functions for real-space excitations. The retarded Greens function in this case is

$$G(\boldsymbol{r}\sigma;t) = -i\theta(t)\left\langle\Psi_{\text{gs}}\right|\{c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t), c^\dagger_{\boldsymbol{r}_d,\sigma}\}\left|\Psi_{\text{gs}}\right\rangle . \tag{5.26}$$

Since our lattice model has discrete translation symmetry, we instead calculate the equivalent quantity

$$G(\boldsymbol{r}\sigma;t) = -i\theta(t)\frac{1}{2}\sum_{\boldsymbol{r}_d}\left[\left\langle\Psi_{\text{gs}}\right|\{c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t), c^\dagger_{\boldsymbol{r}_d,\sigma}\}\left|\Psi_{\text{gs}}\right\rangle + \left\langle\Psi_{\text{gs}}\right|\{c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}, c^\dagger_{\boldsymbol{r}_d,\sigma}(-t)\}\left|\Psi_{\text{gs}}\right\rangle\right] . \tag{5.27}$$

Similar to the $k-$space Greens functions, we consider only the first term for now, and express the groundstates as a superposition of auxiliary model Greens functions:

$$\begin{aligned}\sum_{\boldsymbol{r}_d}\left\langle\Psi_{\text{gs}}\right|c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\Psi_{\text{gs}}\right\rangle &= \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d}\left\langle\psi_{\text{gs}}(\boldsymbol{R}+\boldsymbol{\Delta})\right|c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{R})\right\rangle \\ &= \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta}}\left\langle\psi_{\text{gs}}(-\boldsymbol{r}_d)\right|c_{\boldsymbol{r}-\boldsymbol{R}-\boldsymbol{\Delta},\sigma}(t)T(\boldsymbol{\Delta})c^\dagger_{-\boldsymbol{R},\sigma}\left|\psi_{\text{gs}}(-\boldsymbol{r}_d)\right\rangle .\end{aligned} \tag{5.28}$$

Local Greens function: $\boldsymbol{r} = 0$

We first consider the case of local Greens function, by setting $\boldsymbol{r} = 0$. We also change the dummy variable $\boldsymbol{r}_d$, $\boldsymbol{R}$ and Δ to $-\boldsymbol{r}_d$, $\boldsymbol{R}$ and $-\boldsymbol{\Delta}$ to make the expressions tidier:

$$\sum_{\boldsymbol{r}_d}\left\langle\Psi_{\text{gs}}\right|c_{\boldsymbol{r}_d,\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\Psi_{\text{gs}}\right\rangle = \frac{1}{N}\sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d}\left\langle\psi_{\text{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{R}+\boldsymbol{\Delta},\sigma}(t)T^\dagger(\boldsymbol{\Delta})c^\dagger_{\boldsymbol{R},\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle . \tag{5.29}$$

We split the full summation into four parts, based on whether $\boldsymbol{R} = \boldsymbol{r}_d$ and whether $\Delta + \boldsymbol{R} = \boldsymbol{r}_d$:

$$\begin{aligned}\sum_{\boldsymbol{r}_d}\left\langle\Psi_{\text{gs}}\right|c_{\boldsymbol{r}_d,\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\Psi_{\text{gs}}\right\rangle &= \left\langle\psi_{\text{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{r}_d,\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle \\ &+ \sum_{\boldsymbol{\Delta}\neq 0}\left\langle\psi_{\text{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{r}_d+\boldsymbol{\Delta},\sigma}(t)T^\dagger(\boldsymbol{\Delta})c^\dagger_{\boldsymbol{r}_d,\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle \\ &+ \sum_{\boldsymbol{R}\neq\boldsymbol{r}_d}\left\langle\psi_{\text{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{r}_d,\sigma}(t)T^\dagger(\boldsymbol{r}_d-\boldsymbol{R})c^\dagger_{\boldsymbol{R},\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle \\ &+ \sum_{\boldsymbol{R}\neq\boldsymbol{r}_d,\boldsymbol{\Delta}\neq\boldsymbol{r}_d}\left\langle\psi_{\text{gs}}(\boldsymbol{r}_d)\right|c_{\boldsymbol{\Delta},\sigma}(t)T^\dagger(\boldsymbol{\Delta}-\boldsymbol{R})c^\dagger_{\boldsymbol{R},\sigma}\left|\psi_{\text{gs}}(\boldsymbol{r}_d)\right\rangle ,\end{aligned} \tag{5.30}$$

where we have used the fact that the location of the auxiliary model does not affect the correlation calculated within the auxiliary model in order to cancel out the summation over $\boldsymbol{r}_d$ with the factor of $1/N$. The reason for splitting the expression into these parts is to explicitly write out which operators act on the impurity site $\boldsymbol{r}_d$ and which act on other real-space sites away from the impurity site. We will now replace the real-space excitation operators away from the impurity site with the associated $T-$matrices, following the discussion above eq. 5.11:

$$\mathcal{T}_{\boldsymbol{r},\sigma} = \left[\left(\sum_{\sigma'} c^\dagger_{\boldsymbol{r}_d\sigma} + \text{h.c.}\right) + \left(S^+_{\boldsymbol{r}_d} + \text{h.c.}\right) + \left(C^+_{\boldsymbol{r}_d} + \text{h.c.}\right)\right] c_{\boldsymbol{r}\sigma} \ . \tag{5.31}$$

We also insert a complete basis of states of the auxiliary model at $\boldsymbol{r}_d$, $1 = \sum_n |\psi_n(\boldsymbol{r}_d)\rangle\langle\psi_n(\boldsymbol{r}_d)|$, generating an overlap factor F_n as before:

$$\begin{aligned}\sum_{\boldsymbol{r}_d}\langle\Psi_{\text{gs}}|c_{\boldsymbol{r}_d,\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}|\Psi_{\text{gs}}\rangle &= \langle\psi_{\text{gs}}(\boldsymbol{r}_d)|c_{\boldsymbol{r}_d,\sigma}(t)c^\dagger_{\boldsymbol{r}_d,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \\ &+ \sum_{\boldsymbol{r}_1,\boldsymbol{r}_2\neq r_d,n} F_n(\boldsymbol{r}_1-\boldsymbol{r}_2)^*\langle\psi_{\text{gs}}(\boldsymbol{r}_d)|\mathcal{T}_{\boldsymbol{r}_1,\sigma}(t)|\psi_n(\boldsymbol{r}_d)\rangle\langle\psi_n(\boldsymbol{r}_d)|\mathcal{T}^\dagger_{\boldsymbol{r}_2,\sigma}|\psi_{\text{gs}}(\boldsymbol{r}_d)\rangle \ .\end{aligned} \tag{5.32}$$

Several terms have been dropped because those contributions vanish for systems where the eigenstates conserve total spin S_z.

We next treat the time-evolution of the operators exactly as in the $k-$space Greens function. We replace the one-particle Hamiltonian $H_{1\text{p}}$ with its local component H_{loc} (such as any on-site term of inter-layer hopping, etc). As done previously, we will make use of the eigenstates $\nu(\boldsymbol{r})$ of H_{loc}, with energy $\epsilon^{\text{loc}}_\nu$. Without repeating the details, we state the final result here, for the frequency-domain real-space local Greens function:

$$\begin{aligned}G_{\text{loc}}(\omega) =& \frac{1}{2}\sum_\nu\left[C_{\nu,\boldsymbol{r}_d}\mathcal{G}(\nu(r_d)\sigma, c^\dagger_{r_d,\sigma};\tilde{\omega}) + C^*_{\nu,\boldsymbol{r}_d}\mathcal{G}(c_{r_d,\sigma},\nu^\dagger(r_d)\sigma;\tilde{\omega})\right] \\ &+ \frac{1}{2}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2\neq r_d,\nu} F_n(\boldsymbol{r}_1-\boldsymbol{r}_2)^*\left[C_{\boldsymbol{r}_1,\nu}\mathcal{G}(\mathcal{T}_{\nu(\boldsymbol{r}_1)\sigma},\mathcal{T}^\dagger_{\boldsymbol{r}_2\sigma};\tilde{\omega}) + C^*_{\boldsymbol{r}_1,\nu}\mathcal{G}(\mathcal{T}_{\boldsymbol{r}_1\sigma},\mathcal{T}^\dagger_{\nu(\boldsymbol{r}_2)\sigma};\tilde{\omega})\right] \ ,\end{aligned} \tag{5.33}$$

where the hermitian conjugate in each part comes from considering the second term in eq. 5.27. The shifted frequency $\tilde{\omega} = \omega - \epsilon^{\text{loc}}_\nu$ captures the effect of local terms in $H_{1\text{p}}$; the shift is defined as $e^{i\epsilon^{\text{loc}}_\nu t}c^\dagger_\nu = e^{iH_{\text{loc}}t}c^\dagger_\nu e^{-iH_{\text{loc}}t}$. The operator $\nu^\dagger(\boldsymbol{r})$ represents the eigenstates of H_{loc} at any site $\boldsymbol{r}$. The coefficients $C_{\nu,\boldsymbol{r}}$ express the eigenstates in terms of local states: $c_{\boldsymbol{r}} = \sum_\nu C_{\nu,\boldsymbol{r}}c_{\nu(\boldsymbol{r})}$.

Of course, the states ν will differ from the local states $c_{\boldsymbol{r}}$ only if there are multiple local orbitals (say α) that can hybridise among each other. In such a case, one can also think of an inter-orbital Greens function, defined as

$$G_{\text{loc}}(\alpha,\alpha';t) = -i\theta(t)\langle\Psi_{\text{gs}}|\{c_{\boldsymbol{r}_d,\sigma,\alpha}(t), c^\dagger_{\boldsymbol{r}_d,\sigma,\alpha'}\}|\Psi_{\text{gs}}\rangle \ . \tag{5.34}$$

Such a Greens function can also be calculated from our approach:

$$\begin{aligned}G_{\text{loc}}(\alpha,\alpha';\omega) =& \frac{1}{2}\sum_\nu\left[C_{\nu,\alpha}\mathcal{G}(\nu(r_d,\alpha)\sigma, c^\dagger_{r_d,\sigma,\alpha'};\tilde{\omega}) + C^*_{\nu,\alpha'}\mathcal{G}(c_{r_d,\sigma,\alpha},\nu^\dagger(r_d,\alpha')\sigma;\tilde{\omega})\right] \\ &+ \frac{1}{2}\sum_{\boldsymbol{r}_1,\boldsymbol{r}_2\neq r_d,\nu} F_n(\boldsymbol{r}_1-\boldsymbol{r}_2)^*\left[C_{\alpha,\nu}\mathcal{G}(\mathcal{T}_{\nu(\boldsymbol{r}_1),\sigma,\alpha},\mathcal{T}^\dagger_{\boldsymbol{r}_2,\sigma,\alpha'};\tilde{\omega}) + C^*_{\alpha',\nu}\mathcal{G}(\mathcal{T}_{\boldsymbol{r}_1,\sigma,\alpha},\mathcal{T}^\dagger_{\nu(\boldsymbol{r}_2),\sigma,\alpha'};\tilde{\omega})\right] \ ,\end{aligned} \tag{5.35}$$

Non-local Greens function: $\boldsymbol{r} \neq 0$

We next consider the case of non-local Greens function, where the excitations propagate a non-zero distance $\boldsymbol{r}$:

$$\sum_{\boldsymbol{r}_d} \langle \Psi_{\text{gs}} | c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t) c^\dagger_{\boldsymbol{r}_d,\sigma} | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{R}+\boldsymbol{\Delta}+\boldsymbol{r},\sigma}(t) T^\dagger(\boldsymbol{\Delta}) c^\dagger_{\boldsymbol{R},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ . \tag{5.36}$$

To bring this expression closer to the one for the local Greens function, we redefine $\Delta + \boldsymbol{r} \to \Delta$:

$$\sum_{\boldsymbol{r}_d} \langle \Psi_{\text{gs}} | c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t) c^\dagger_{\boldsymbol{r}_d,\sigma} | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\boldsymbol{R},\boldsymbol{\Delta},\boldsymbol{r}_d} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{R}+\boldsymbol{\Delta},\sigma}(t) T^\dagger(\boldsymbol{\Delta}) T(\boldsymbol{r}) c^\dagger_{\boldsymbol{R},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ . \tag{5.37}$$

This allows to again split the full summation based on whether $\boldsymbol{R} = \boldsymbol{r}_d$ and whether $\Delta + \boldsymbol{R} = \boldsymbol{r}_d$:

$$\begin{aligned}
\sum_{\boldsymbol{r}_d} \langle \Psi_{\text{gs}} | c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t) c^\dagger_{\boldsymbol{r}_d,\sigma} | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) T(\boldsymbol{r}) c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\
&+ \sum_{\boldsymbol{\Delta}\neq 0} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d+\boldsymbol{\Delta},\sigma}(t) T^\dagger(\boldsymbol{\Delta} - \boldsymbol{r}) c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\
&+ \sum_{\boldsymbol{R}\neq \boldsymbol{r}_d} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) T^\dagger(\boldsymbol{r}_d - \boldsymbol{R} - \boldsymbol{r}) c^\dagger_{\boldsymbol{R},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\
&+ \sum_{\boldsymbol{R}\neq \boldsymbol{r}_d, \boldsymbol{\Delta}\neq \boldsymbol{r}_d} \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{\Delta},\sigma}(t) T^\dagger(\boldsymbol{\Delta} - \boldsymbol{R} - \boldsymbol{r}) c^\dagger_{\boldsymbol{R},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ .
\end{aligned} \tag{5.38}$$

Introducing Fourier transforms of the operators and inserting eigenstates of the auxiliary models leads to

$$\begin{aligned}
&\sum_{\boldsymbol{r}_d} \langle \Psi_{\text{gs}} | c_{\boldsymbol{r}_d+\boldsymbol{r},\sigma}(t) c^\dagger_{\boldsymbol{r}_d,\sigma} | \Psi_{\text{gs}} \rangle \\
&= \sum_n F_n^*(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\
&+ \sum_{\boldsymbol{k},n} F_n'(\boldsymbol{k}) F_n^*(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{\boldsymbol{k},\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\
&+ \sum_{\boldsymbol{k},n} F_n'(\boldsymbol{k})^* F_n^*(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{\boldsymbol{k},\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\
&+ \sum_{\boldsymbol{k}_1,\boldsymbol{k}_2} F_n'(\boldsymbol{k}_1) F_n'(\boldsymbol{k}_2)^* F_n^*(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{\boldsymbol{k}_1,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{\boldsymbol{k}_2,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ ,
\end{aligned} \tag{5.39}$$

The net difference from the diagonal Greens function is the presence of the overlap factor $F_n^*(\boldsymbol{r})$.

1.5.3 Equal-Time Ground State Correlators

Apart from dynamical correlations such as Greens functions, we would also like to use our approach to calculate static correlation functions in terms of auxiliary model quantities. We first consider a real-space correlation that tracks the propagation of a general excitation described by the operator $O(\boldsymbol{r})$ over a distance $\boldsymbol{\Delta}$:

$$C_O(\boldsymbol{r}) = \langle \Psi_{\text{gs}} | O(\boldsymbol{r}_d + \boldsymbol{r}) O^\dagger(\boldsymbol{r}_d) | \Psi_{\text{gs}} \rangle \ . \tag{5.40}$$

We first consider a local correlation: $\boldsymbol{r} = 0$. To obtain a tractable expression for this, we note that the quantity being computed here is almost identical to the one computed in eq. 5.28, the only difference being the time-evolution of the operator. For the local correlation, we therefore assume the result of eq. 5.31, dropping the time evolution because the correlators in this section are calculated at equal time. We also transform from $k-$space back to real-space because there are no time-evolution operators which need extended states in order to be resolved. Putting all this together, we get

$$\begin{aligned} C_O^{\text{loc}} &= \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | O(\boldsymbol{r}_d) O^\dagger(\boldsymbol{r}_d) | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &\quad + \sum_{\boldsymbol{r} \neq 0, n} F_n(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{r}_d)} | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | O^\dagger(\boldsymbol{r}_d) | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &\quad + \sum_{\boldsymbol{r} \neq 0, n} F_n(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | O(\boldsymbol{r} + \boldsymbol{r}_d) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{O(\boldsymbol{r}_d)} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &\quad + \sum_{\boldsymbol{r}_1 \neq 0, \boldsymbol{r}_2 \neq 0, n} F_n(\boldsymbol{r}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{O(\boldsymbol{r}_1+\boldsymbol{r}_d)} | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{O(\boldsymbol{r}_2+\boldsymbol{r}_d)} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ , \end{aligned} \tag{5.41}$$

where the transition operator $\mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{r}_d)}$ represents impurity-bath correlation:

$$\mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{r}_d),\sigma} = \left[c^\dagger_{\boldsymbol{r}_d \sigma} + S^\sigma_{\boldsymbol{r}_d} + C^+_{\boldsymbol{r}_d} \right] O(\boldsymbol{r} + \boldsymbol{r}_d) \ , \tag{5.42}$$

We next consider non-local correlations over a distance $\boldsymbol{\Delta}$. Adapting eq. 5.39 for the non-local real-space Greens function in the same manner, we get

$$\begin{aligned} C_O(\boldsymbol{\Delta}) &= \sum_n F_n^*(\boldsymbol{\Delta}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | c_{\boldsymbol{r}_d,\sigma}(t) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | c^\dagger_{\boldsymbol{r}_d,\sigma} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &\quad + \sum_{\boldsymbol{r} \neq 0, n} F_n(\boldsymbol{r} + \boldsymbol{\Delta}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{O(\boldsymbol{r}+\boldsymbol{\Delta}+\boldsymbol{r}_d)} | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | O^\dagger(\boldsymbol{r}_d) | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &\quad + \sum_{\boldsymbol{r} \neq 0, n} F_n(\boldsymbol{r} + \boldsymbol{\Delta}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | O(\boldsymbol{r} + \boldsymbol{\Delta} + \boldsymbol{r}_d) | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{O(\boldsymbol{r}_d)} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \\ &\quad + \sum_{\boldsymbol{r}_1 \neq 0, \boldsymbol{r}_2 \neq 0, n} F_n(\boldsymbol{r} + \boldsymbol{\Delta}) \, \langle \psi_{\text{gs}}(\boldsymbol{r}_d) | \mathcal{T}_{O(\boldsymbol{r}_1+\boldsymbol{\Delta}+\boldsymbol{r}_d)} | \psi_n(\boldsymbol{r}_d) \rangle \, \langle \psi_n(\boldsymbol{r}_d) | \mathcal{T}^\dagger_{O(\boldsymbol{r}_2+\boldsymbol{r}_d)} | \psi_{\text{gs}}(\boldsymbol{r}_d) \rangle \ , \end{aligned} \tag{5.43}$$

1.6 Entanglement Measures

We will now describe the prescription of calculating entanglement measures of the lattice model from within our auxiliary model treatment. In this section, we are interested mainly in two such measures, the entanglement entropy and the mutual information. Given a pure state $|\Psi\rangle$ describing the complete system, the entanglement entropy $S_{\text{EE}}(\nu)$ of a subsystem ν quantifies the entanglement of ν with the rest of the subsystem, and is defined as

$$S_{\text{EE}}(\nu) = -\text{Tr}\left[\rho(\nu)\log\rho(\nu)\right], \quad \rho(\nu) = \text{Tr}_{\nu}\left[\left|\Psi\right\rangle\left\langle\Psi\right|\right] \tag{6.1}$$

where $\text{Tr}\left[\cdot\right]$ is the trace operator, and $\rho(\nu)$ is the reduced density matrix (RDM) for the subsystem ν obtained by taking the partial trace Tr_{ν} (over the states of ν) of the full density matrix $\rho = \left|\Psi\right\rangle\left\langle\Psi\right|$. If the subsystem ν describes local regions in real space (or states in $k-$space), we might be interested in the entanglement between two such subsystems ν_1 and ν_2. The correct measure to quantify such entanglement is the mutual information:

$$I_2(\nu_1, \nu_2) = S_{\text{EE}}(\nu_1) + S_{\text{EE}}(\nu_2) - S_{\text{EE}}(\nu_1 \cup \nu_2), \tag{6.2}$$

where $\nu_1 \cup \nu_2$ is a larger subsystem formed by combining ν_1 and ν_2.

We start by recalling the decomposition of entanglement entropy in terms of multipartite information as laid out in Subsection **??**. Given a partitioning of the system into mutually exclusive subsystems $\{\nu_i\}$, the average entanglement entropy $S^{(1)}_{\text{EE}}$ over all subsystems can be expressed as

$$\begin{aligned} S^{(1)}_{\text{EE}} &= \frac{1}{N}\sum_{n=2}^{N}(-1)^n\binom{N}{n}\bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!}\sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n), \end{aligned} \tag{6.3}$$

where $\bar{I}_n$ is the average multipartite information I_n over all distinct combinations of n subsystems.

We first take a look at real-space entanglement. Real-space entanglement measures can be used to probe delocalisation-localisation transitions. The simplest such measure is the entanglement of a lattice site in real-space. Since the local entanglement entropy will be uniform at each lattice site for a system with translation invariance, it suffices to calculate the real-space averaged entanglement entropy $S^{(1)}_{\text{EE}}$. The corresponding multipartite measures $\bar{I}_n$ involve collections of lattice sites. We write such a measure as

$$\bar{I}_n = \frac{(N-n)!n!}{N!}\frac{1}{N}\sum_{I(\boldsymbol{r}_d)}\sum_{B^{n-1}(\boldsymbol{r}_d)} I_n(I(\boldsymbol{r}_d); B^{n-1}(\boldsymbol{r}_d)), \tag{6.4}$$

where $I(\boldsymbol{r}_d)$ sums over all lattice sites in the form of an impurity site and $B^{n-1}(\boldsymbol{r}_d)$ considers all possible subsets of size $n-1$ of the remaining bath sites. Together, this choice corresponds to an auxiliary model with the impurity placed at $\boldsymbol{r}_d$. The additional factor of $1/N$ takes care of the overcounting arising from each choice appearing in N auxiliary models. We can therefore write the multipartite information as an average over auxiliary models, each auxiliary model contributing an average I_n over its bath sites:

$$\bar{I}_n = \frac{n}{N^2}\sum_{\boldsymbol{r}_d} I^{(\text{tot})}_n(\boldsymbol{r}_d). \tag{6.5}$$

The quantity $I^{(\text{tot})}_n(\boldsymbol{r}_d)$ is defined as $I^{(\text{tot})}_n(\boldsymbol{r}_d) = \frac{(N-n)!(n-1)!}{(N-1)!}\sum_{B^{n-1}(\boldsymbol{r}_d)} I_n(I(\boldsymbol{r}_d); B^{n-1}(\boldsymbol{r}_d))$. By translation invariance, such a measure will be uniform across auxiliary models and it suffices to calculate the measure for a single auxiliary model located at a reference point $\boldsymbol{r}_d$; the whole expression therefore simplifies to

$$S_{\rm EE}^{(1)} = \frac{1}{N}\sum_{n=2}^{N}(-1)^n\binom{N-1}{n-1}I_n^{(\rm tot)}(\boldsymbol{r}_d) = \frac{1}{N}\sum_{n=2}^{N}(-1)^n\sum_{B^{n-1}(\boldsymbol{r}_d)} I_n(I(\boldsymbol{r}_d);B^{n-1}(\boldsymbol{r}_d))\ . \tag{6.6}$$

For example, truncating the summation to $n = 3$, we have the explicit expression

$$S_{\rm EE}^{(1)} = \frac{N-1}{N}I_2^{(\rm tot)}(\boldsymbol{r}_d) - \frac{(N-1)(N-2)}{2N}I_3^{(\rm tot)}(\boldsymbol{r}_d) \tag{6.7}$$

Chapter 2

Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation

2.1 Bilayer extended Hubbard model: Impurity model

[10] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{1\text{p}} + H_f + H_d + H_{fd} \ , \tag{1.1}$$

where $H_{1\text{p}}$ is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{1\text{p}} = -t \sum_{\sigma,\alpha} \sum_{\langle i,j \rangle} \left(c^\dagger_{i,\sigma,\alpha} c_{j,\sigma,\alpha} + \text{h.c.} \right) - \sum_{i,\sigma,\alpha} \mu_\alpha n_{i,\sigma,\alpha} \ , \tag{1.2}$$

such that α sums over the two layers f and d. H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = -\frac{U_f}{2}(n_{f,\uparrow} - n_{f,\downarrow})^2 + \sum_{Z \in \text{NN}} \Bigg[-V \sum_\sigma \left(c^\dagger_{f,\sigma} c_{Z,\sigma,f} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,f} c_{Z,\beta,f} \\ - \frac{W_f}{2} \left(n_{Z,\uparrow,f} - n_{Z,\downarrow,f} \right)^2 \Bigg] \ , \\ H_d = -\eta_d n_d - \frac{U_d}{2}(n_{d,\uparrow} - n_{d,\downarrow})^2 + \sum_{Z \in \text{NN}} \Bigg[-V \sum_\sigma \left(c^\dagger_{d,\sigma} c_{Z,\sigma,d} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha,\beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\alpha,d} c_{Z,\beta,d} \Bigg] \ , \end{aligned} \tag{1.3}$$

where $c^\dagger_{\nu,\sigma,\alpha}$ can refer to creation operator for either the $f-$layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp \sum_\sigma \left(c^\dagger_{d,\sigma} c_{f,\sigma} + \text{h.c.} \right) \ . \tag{1.4}$$

2.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the "unit" cell of translation. As discussed in the text surrounding eq. 3.9, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. Placing the impurity sites at $\boldsymbol{r}_d$ in each layer, we get

$$H_{\text{clust}}(\boldsymbol{r}_d) = H_f(\boldsymbol{r}_d) + H_d(\boldsymbol{r}_d) + H_{fc}(\boldsymbol{r}_d) \ . \tag{2.1}$$

The details of tiling a cluster Hamiltonian were already shown in Sec. 1.3.2; the general idea is to use manybody translation operators (details in Sec. 1.3.1) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_r T^\dagger(\boldsymbol{r}) H_{\text{clust}}(\boldsymbol{r}_d) T(\boldsymbol{r}) \\ &= -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.}\right) - \sum_i \left[\eta_d n_{i,d} + \frac{U_d}{2}(n_{i,\uparrow,d} - n_{i,\downarrow,d})^2\right] + \frac{2J}{Z}\sum_{\langle i,j\rangle} \boldsymbol{S}_{i,d}\cdot \boldsymbol{S}_{j,d} \\ &\quad - \frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - \frac{U_f - W_f}{2}\sum_i \left(n_{i,\uparrow,f} - n_{i,\downarrow,f}\right)^2 + \frac{2J}{Z}\sum_{\langle i,j\rangle} \boldsymbol{S}_{i,f}\cdot \boldsymbol{S}_{j,f} \\ &\quad - V_\perp \sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) \end{aligned} \tag{2.2}$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the $d-$layer; a non-zero value moves the layer away from half-filling. The $f-$layer is pinned at half-filling.

Following Sec. 1.5.1, the retarded $k-$space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\boldsymbol{k}, \sigma, \alpha; \omega) = \sum_\nu C_{\nu,\boldsymbol{k}\alpha}\mathcal{G}(\mathcal{T}_{\nu,\sigma,\alpha}; \omega - \mathcal{E}_\nu(\boldsymbol{k})) \tag{2.3}$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian $H_{1\text{p}}$, with energies $\mathcal{E}_\nu(\boldsymbol{k})$:

$$H_{1\text{p}} = -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.} + f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} \ , \tag{2.4}$$

and the $\mathcal{T}-$matrices are defined in eq. 5.11:

$$\mathcal{T}_{\nu,\sigma,\alpha} = \left[\left(\sum_{\sigma'} c^\dagger_{\alpha\sigma} + \text{h.c.}\right) + \left(S^+_\alpha + \text{h.c.}\right) + \left(C^+_\alpha + \text{h.c.}\right)\right] c_{\nu,\sigma} \ , \tag{2.5}$$

In order to calculate $\mathcal{E}_\nu(\boldsymbol{k})$, we introduce the $k-$space modes via a Fourier transform:

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} c^\dagger_{\boldsymbol{k},\sigma}\ , \quad f^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i} f^\dagger_{\boldsymbol{k},\sigma}\ . \tag{2.6}$$

In terms of these modes, the 1−particle Hamiltonian decouples in $k-$space:

$$H_{1\text{p}} = \sum_{\boldsymbol{k},\sigma}\left[(\epsilon_{\boldsymbol{k}} - \eta_d)(n_{\boldsymbol{k},\sigma,d} + n_{\boldsymbol{k},\sigma,f}) - V_\perp(c^\dagger_{\boldsymbol{k},\sigma} f_{\boldsymbol{k},\sigma} + \text{h.c.})\right]\ . \tag{2.7}$$

The diagonal modes are given by

$$\nu^\dagger_{\boldsymbol{k},\sigma,\pm} = C_{\pm,k} c^\dagger_{\boldsymbol{k},\sigma} + \sqrt{1 - C^2_{\pm,k}} f^\dagger_{\boldsymbol{k},\sigma}\ , \tag{2.8}$$

the coefficients obtained from diagonalising the matrix

$$H_{1\text{p}}(\boldsymbol{k}) = \begin{pmatrix} \epsilon_{\boldsymbol{k}} & -V_\perp \\ -V_\perp & \epsilon_{\boldsymbol{k}} - \eta_d \end{pmatrix}\ . \tag{2.9}$$

The energies are given by $\mathcal{E}_\pm(\boldsymbol{k}) = \epsilon_k - \eta_d/2 \pm \Delta$, with ϵ_k being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2/4}$. Substituting into the expression for the lattice Greens function, we get

$$\begin{aligned} G(\boldsymbol{k}, d; \omega) =& C_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k}+,d}, \mathcal{T}^\dagger_{\boldsymbol{k},d}; \omega - \mathcal{E}_+(\boldsymbol{k})) + C_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k}-,d}, \mathcal{T}^\dagger_{\boldsymbol{k},d}; \omega - \mathcal{E}_-(\boldsymbol{k})) \\ &+ C^*_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},d}, \mathcal{T}^\dagger_{\boldsymbol{k}+,d}; \omega - \mathcal{E}_+(\boldsymbol{k})) + C^*_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},d}, \mathcal{T}^\dagger_{\boldsymbol{k}-,d}; \omega - \mathcal{E}_-(\boldsymbol{k}))\ . \end{aligned} \tag{2.10}$$

Following eq. 5.33, we can similarly write down an expression the local lattice Greens function for the $f-$ and $d-$electrons. For that, we define the local diagonal modes $\nu_\pm$ obtained from the local Hamiltonian

$$H_{\text{loc}} = \begin{pmatrix} 0 & -V_\perp \\ -V_\perp & 0 - \eta_d \end{pmatrix}\ , \tag{2.11}$$

and their associated energies $\varepsilon_\pm$. Within the impurity model, these local bonding/antibonding modes can be constructed on the impurity sites ($\nu_\pm$) or on the neighbouring bath sites, ($\nu^Z_\pm$). The local Greens function can be expressed in terms of these modes:

$$\begin{aligned} G_{dd}(\omega) =& \frac{1}{2}\sum_\nu C_{\nu,d}\left[\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + \mathcal{G}(d\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z} C_{\nu^Z,Z_d}\left[\mathcal{G}(\mathcal{T}_{Z_d,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z}) + \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z})\right]\ . \end{aligned} \tag{2.12}$$

The presence of local hybridisation between the $d-$ and $f-$ orbitals makes the inter-orbital Greens function important as well. Following eq. 5.35, we have

$$\begin{aligned} G_{fd}(\omega) =& \frac{1}{2}\sum_\nu \left[C_{\nu,f}\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + C_{\nu,d}\mathcal{G}(f\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z}\left[C_{\nu^Z,Z_f}\mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z}) + C_{\nu^Z,Z_d}\mathcal{G}(\mathcal{T}_{Z_f,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z})\right]\ . \end{aligned} \tag{2.13}$$

Using these, we can construct the hybridised Greens function associated with the states ν:

$$G_{\nu_+}(\omega) = C_{+,d}^2 G_{dd}(\omega) + C_{+,f}^2 G_{ff}(\omega) + C_{+,d}C_{+,f}\left[G_{df}(\omega) + G_{fd}(\omega)\right] \ , \tag{2.14}$$

2.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ES-IAM

In the limit of large U_α, we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\text{aux}} = \sum_{\mathbf{k},\sigma,\alpha} \epsilon_{\mathbf{k},\alpha} n_{\mathbf{k},\sigma,\alpha} + \sum_\alpha \sum_{Z\in\text{NN}} \left[\frac{1}{2}J_\alpha \sum_{\sigma,\sigma'} \mathbf{S}_\alpha \cdot \boldsymbol{\sigma}_{\alpha\beta} c^\dagger_{Z,\sigma,\alpha} c_{Z,\sigma',\alpha} - \frac{W_\alpha}{2}\left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha}\right)^2\right] + J\mathbf{S}_f \cdot \mathbf{S}_d \ . \tag{3.1}$$

The Kondo coupling and bath interaction acquire $k-$dependence:

$$\begin{aligned} J(\mathbf{k},\mathbf{q}) &= \frac{J}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)\right] \ , \\ W(\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}') &= \frac{W}{2}\left[\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)\right] \ . \end{aligned} \tag{3.2}$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi,\pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi,-\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q},\sigma,f\rangle$ and $|\mathbf{q},\sigma,d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}},\sigma,f\rangle$ and $|\bar{\mathbf{q}},\sigma,d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration) For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states. We already have the renormalisation group equations for the couplings J_α in the case of $J = 0$:

$$\frac{\Delta J_f(\mathbf{k}_1,\mathbf{k}_2)}{\Delta D} = -\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[J_f(\mathbf{k}_1,\mathbf{q})J_f(\mathbf{q},\mathbf{k}_2) + 4J_f(\mathbf{q},\bar{\mathbf{q}})W_f(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)\right] G_f^{(0)}(\mathbf{q}) \ , \tag{3.3}$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G_f^{(0)}$ is the propagator for the excited intermediate state (for $J = 0$):

$$G_f^{(0)}(\mathbf{q}) = \frac{1}{\omega - |\epsilon_f(\mathbf{q})|/2 + J_f(\mathbf{q})/4 + W_f(\mathbf{q})/2} \ . \tag{3.4}$$

An identical equations holds for ΔJ_d, with al $f-$quantities replaced with $d-$quantities. Switching on J modifies the excitation energy in the propagator:

$$G_f(\mathbf{q}) = \frac{1}{1/G_f^{(0)} - J\vec{S}_f \cdot \vec{S}_d} \ . \tag{3.5}$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4}\mathcal{P}_0 + \frac{1}{4}\mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I} \,. \tag{3.6}$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^{(0)} + 3J/4}\mathcal{P}_0 + \frac{1}{1/G_f^{(0)} - J/4}\mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^{(0)} + 3J/4} + \frac{3/4}{1/G_f^{(0)} - J/4}\right)\mathbb{I} + \left(\frac{1}{1/G_f^{(0)} + 3J/4} + \frac{1}{1/G_f^{(0)} - J/4}\right)\vec{S}_f \cdot \vec{S}_d \,. \end{aligned} \tag{3.7}$$

Noting that only the first operator renormalises the Kondo interaction, we get the following modified RG equation for the Kondo coupling J_α:

$$\begin{aligned} \frac{\Delta J_\alpha}{\Delta D} = \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_\alpha(\mathbf{k}_1, \mathbf{q})J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}})W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)\right] \Bigg(\frac{1}{4}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + 3J/4} \\ + \frac{3}{4}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - J/4}\Bigg) \,, \end{aligned} \tag{3.8}$$

We now turn to the renormalisation of J, arising from hybridisation of $f-$ and $d-$layers. In order to capture coherent scattering processes between the two layers, we project onto the following rotated basis of excited states:

$$\begin{aligned} |H\rangle_\pm &= \frac{1}{\sqrt{2}}\left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d\right) \,, \\ \mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- \,, \end{aligned} \tag{3.9}$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α: $|H(q)\rangle_\alpha = c^\dagger_{\mathbf{q},\alpha} c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $JS_f^+S_d^-$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the $d-$layer followed by a spin-flip of the $f-$layer:

$$\Delta H = \frac{1}{N^2}\sum_{\mathbf{q}\in\mathrm{UV}} \langle L|\frac{1}{2}J_f(\bar{\mathbf{q}}, \mathbf{q})S_f^+ c^\dagger_{\mathbf{q}-\downarrow,f} c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2}J_d(\mathbf{q}, \bar{\mathbf{q}})S_d^- c^\dagger_{\mathbf{q}\uparrow,d} c_{\bar{\mathbf{q}}\downarrow,d}|L\rangle \,, \tag{3.10}$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D} \,, \tag{3.11}$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q},\sigma\in\mathbf{q}_\pm,\alpha} \epsilon_\alpha(\mathbf{q})\tau_{\mathbf{q},\sigma} + \frac{1}{2}\sum_\alpha J_\alpha S^z_\alpha \sum_{\mathbf{q},\sigma} \sigma n_{\mathbf{q},\sigma,\alpha} + J\vec{S}_f\cdot\vec{S}_d \ . \tag{3.12}$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H = \frac{1}{2}\left(G_f + G_d\right)\left(|H\rangle_+\langle H|_+ + |H\rangle_-\langle H|_-\right) + \frac{1}{2}\left(G_f - G_d\right)\left(|H\rangle_+\langle H|_- + |H\rangle_-\langle H|_+\right) \ , \tag{3.13}$$

where

$$G_\alpha(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q}) - \frac{1}{4}J} \ . \tag{3.14}$$

In eq. 3.10, since the first process is an excitation into a $d-$state and the second process is a de-excitation from an $f-$state, the expression can be simplified into

$$\begin{aligned}\langle L|S^+_f c^\dagger_{\mathbf{q}-\downarrow,f}c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H S^-_d c^\dagger_{\mathbf{q}+\uparrow,d}c_{\mathbf{q}-\downarrow,d}|L\rangle &= S^+_f S^-_d \langle H_f|\mathcal{P}(\mathbf{q})_H G\mathcal{P}(\mathbf{q})_H|H_d\rangle \\ &= \frac{1}{2}\left(G_f + G_d\right)S^+_f S^-_d \ .\end{aligned} \tag{3.15}$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2}\sum_{\mathbf{q}\in\text{UV}} J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right)S^+_f S^-_d \ . \tag{3.16}$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N}S^+_f S^-_d \int_{UV} d\mathbf{q}\,\rho(\mathbf{q})J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) \ . \tag{3.17}$$

The time-reversed of this process can be obtained simply by exchanging the $d-$interaction with the $f-$interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S^+_f S^-_d$ is therefore

$$\begin{aligned}\frac{\Delta J}{\Delta D} &= \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})\left(G_f(\mathbf{q}) + G_d(\mathbf{q})\right) \\ &= \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\left[\frac{J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right] \ ,\end{aligned} \tag{3.18}$$

where $\bar{\epsilon}_\alpha(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q},\bar{\mathbf{q}}$.

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = -J_\alpha\mathcal{G}_\alpha J^\dagger_\alpha - 4\mathcal{W}_\alpha\text{Tr}\left[\mathcal{G}'_\alpha\right] \ , \tag{3.19}$$

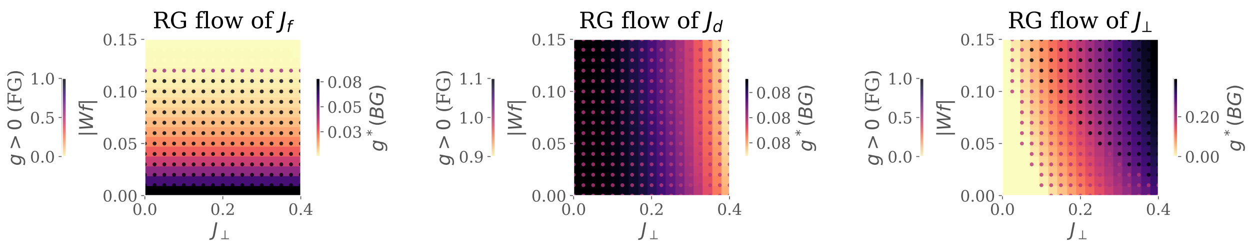

Figure 2.1: RG phase diagrams for various Kondo couplings: $f-$layer (left), $d-$layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the $f-$layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, $1/2$ or 0 respectively.

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q,q] = \begin{cases} \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2-J/4}\right), & \mathbf{q}\in \text{PS}\ , \\ \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2-J/4}\right), & \mathbf{q}\in \text{HS}\ , \end{cases} \tag{3.20}$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k,k'] = \frac{W}{2}\left[\cos(k_x - k_x') + \cos(k_y - k_y')\right]$. Finally, $\mathcal{G}_\alpha'$ is another diagonal matrix whole matrix elements are defined as $\mathcal{G}_\alpha'[q,q] = \mathcal{G}[q,q]J_\alpha[q,\bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2}\mathbf{J}\cdot\mathbf{G}\ , \tag{3.21}$$

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q})\left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right]. \tag{3.22}$$

2.4 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 2.1 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the $f-$layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the $d-$layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the $f-$layer becomes RG-irrelevant for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the $f-$impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong $J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the $f-$layer hybridise with the itinerant electrons of the $d-$layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the $f-$layer forms local moments which decouple from the $d-$layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the $f-$electrons remain itinerant and it is they who hybridise with the $d-$electrons to again form two bands separated by a hybridisation gap.

2.5 Results at half-filling

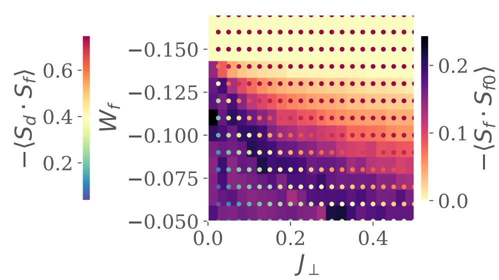

Figure 2.2: *Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the $f-$impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the $f-$impurity and the $d-$impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:*

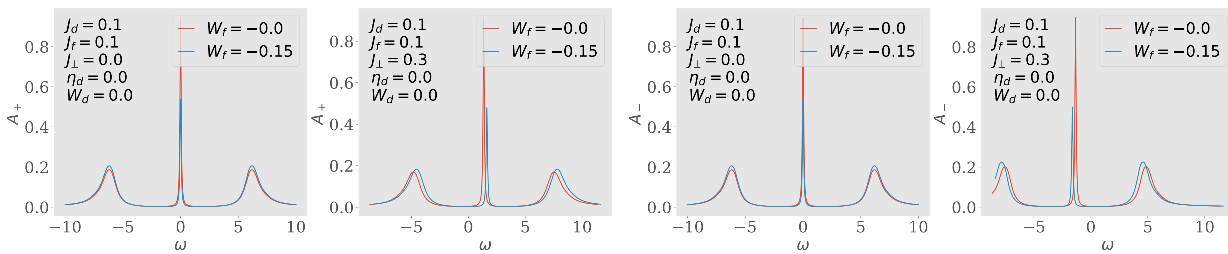

Figure 2.3: *Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function $G_{k,\pm} = G(k_d \pm k_f; \omega)$. There is no hybridisation for $J_\perp = 0$ (first and third plots), and the only effect of W_f is to shrink the height of the central resonance. Introducing a non-zero $J_\perp$ splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where $\omega = 0$ resides inside the hybridisation gap.*

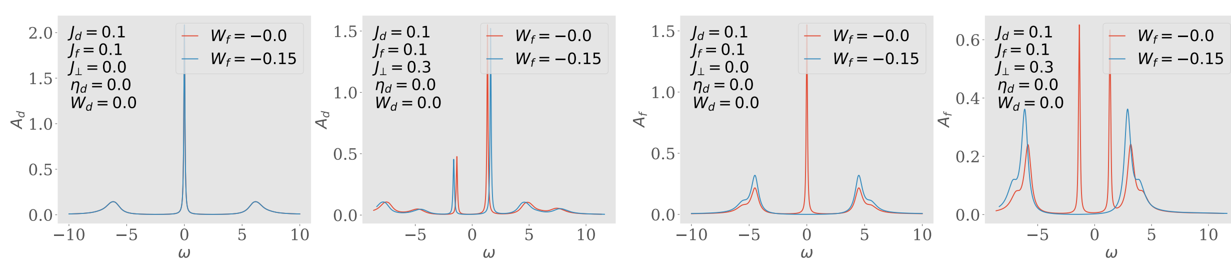

Figure 2.4: *Spectral functions A_d and A_f of the auxiliary model, for excitations in the $d-$ and $f-$layers respectively. Both show a central Kondo resonance for $J_\perp = 0$ which splits into two bands upon adding a non-zero $J_\perp$. Increasing $|W_f|$ to large values gaps out the local Fermi liquid excitations of A_f.*

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